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Latent Commonality Expectation-Maximisation for Box-supervised Tree Crown Instance Segmentation

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Abstract: Individual tree crown segmentation from aerial imagery underpins tree-level carbon accounting, biodiversity, and restoration monitoring at landscape scale. However, existing models are predominantly trained on dense canopy forest imagery and degrade in savannah and drylands, where tree crowns are sparse, of variable appearance, and underrepresented in annotated benchmarks. These models also typically depend on costly polygon annotations. We introduce LACE (LAtent Commonality Expectation-maximisation), a box-supervised instance segmentation model, evaluated on 0.1 m/px aerial RGB tree crown imagery. LACE uses a frozen DINOv3-web ViT-L/16 encoder, applied at four spatial offsets and interlaced into a denser feature grid, with a lightweight CenterNet-style detection head trained solely on bounding boxes. We use expectation-maximisation to separate recurring appearance, the "treeness", within bounding boxes from surroundings. On the OAM-TCD benchmark test set, LACE reaches a mask AP_{50} of 0.647 ± 0.003 (3 seeds) trained on 900 box-annotated images and without mask annotations, level with the 0.648 of Restor's released mask-supervised Mask R-CNN, which was trained on the full $\sim$ 4.2k image set with polygon masks. On a sparse-canopy holdout set, mask AP_{50} rises to 0.678 versus 0.625 for Detectree2, a mask-supervised baseline. On NeonTreeEvaluation, using the official evaluation code, LACE reaches 0.728 ± 0.003 F1@0.4 (5 seeds) from 23,424 hand-annotated RGB boxes alone, matching the authors' DeepForest model's published 0.719, using under 0.1% of its training annotations and none of its LiDAR-derived 30M-crown pretraining set. By leveraging frozen self-supervised features, LACE matches fully-supervised specialist baselines from boxes alone, removing the need for polygon annotation in tree crown instance segmentation for sparse-canopy environments where labelled data is scarce.

Keywords: Tree crown instance segmentation, self-supervised vision transformers, remote sensing, carbon accounting, DINOv3, expectation-maximisation, box-supervised instance segmentation

1. Introduction

Accurate measurement of individual tree crowns can serve as a proxy for woody biomass estimation via allometries [1,2] and thereby enable monitoring of carbon sequestration within areas containing trees since carbon content can be derived from woody biomass. Measurement techniques in this domain have traditionally focused on dense canopy areas and forests, but there is an increasing body of literature that highlights the limitations of this approach as distribution, density, cover, and carbon content of trees in sparse-canopy biomes (listed in Table 6) are not well understood at sub-continental to continental scales [2–4]. Recent studies highlight that the global contribution of sparse-canopy biomes as a carbon sink is underestimated due to the underestimation of tree mass in sparse-canopy areas [2,5–7]. At the same time, 1/4 of all trees globally are located in grassland, savanna, desert and tundra biomes [8], and 29% of Africa's tree cover lies outside areas classified as forest [9], rendering further exploration of these landscapes meaningful and worthwhile.

Several conventional methodologies have been applied to the measurement of tree cover and biomass, including field inventories of plots [10], statistical sampling [11] and

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Figure 1. Crown masks on six windows of the sparse-canopy holdout (1024 px).

hybrid solutions pairing plots with Remote Sensing (RS) covariates such as optical, multispectral, LiDAR and Airborne Laser Scanning (ALS) [12–14]. Machine learning and deep learning applied to satellite and UAV data have since shifted the field from aggregate cover fractions toward mapping individual trees: Brandt et al. [6] mapped over 1.8 billion crowns across 1.3 million km^2 of the Sahara and Sahel from 0.5 m imagery. Tucker et al. [2] extended this to carbon estimation over 9.9 billion individual trees. Mugabowindekwe et al. [15] produced nation-wide tree-level carbon estimates for Rwanda using 0.25 m/px aerial RGB imagery and found that 48.6% of Rwanda's aboveground carbon stock in trees was located outside forests. They also found that "even very detailed manual *forest* delineation…missed 38.4% of the isolated trees in Rwanda, which account for 25.5% of the national aboveground carbon stocks" (emphasis added). These studies establish that *individual* crown detection in open, scattered-tree environments is tractable, but, crucially, typically depend on a) (often proprietary) sub-metre-resolution imagery and b) labelled data, motivating more transferable and label-efficient methods.

Modern deep learning approaches focus on instance segmentation on RGB imagery (Table 1 summarises representative methods): DeepForest [16] established a RetinaNet-based baseline evaluated on the NeonTreeEvaluation benchmark [17], while Detectree2 [18] applied Mask R-CNN [19] to tropical crown polygons. Tong & Zhang [20] applied StarDist [21], a U-Net [22] backbone model applying star-convex polygons as proposals during the detection stage rather than the more commonly used axis-aligned bounding boxes, allowing for improved Non-Maximum Suppression (NMS) performance and more accurate detections. This approach was designed as a solution to the challenging problem of individual tree crown instance segmentation in dense forest canopy, where crowns are often touching and overlapping, which can contribute to many valid detections being needlessly dropped due to the coarse tessellation of bounding box rectangles. Applying a different approach to a similar dense-canopy problem domain, the authors of the tropical rainforest SelvaMask dataset [23] utilised a DINO [24] (a DETR-family detector, unrelated to the self-supervised DINO backbones used here) based tree crown detection model coupled with a Segment Anything Model (SAM) [25] for prediction mask generation, applied to instance segmen-

tation in dense canopy rainforests. Similar concepts of utilising SAM for RS tasks as a second-stage mask generator by providing it with prompts underpins models such as RSPrompter [26] and BalSAM [27], the latter both with and without the integration of a Digital Surface Model (DSM) depth map alongside RGB data.

We aim to address the tree crown instance segmentation space from two angles underexplored in the literature. Figure 1 previews the setting: individual crowns in savanna, scrub and other open landscapes, where trees are sparse, small and set against bare or grassy ground. Firstly, existing approaches focus on dense-canopy forests, and overwhelmingly exclude biomes such as savannah and dryland. For example, the authors of Detectree2 deliberately exclude images with $\leq$ 40% tree canopy cover from their datasets [18]. In the DeepForest paper, Weinstein et al. highlight Onaqui, Utah (ONAQ) as the model's worst-performing site since it is a "desert scrub site with a different vegetation structure from any of the training data" [16]. Secondly, in order to produce output individual tree crown instance polygon masks, existing models typically rely on mask annotations for training [18,28–32], which need to be produced manually by human annotators.

1.1. Lack of sparse-canopy datasets

We recognise a number of contributing factors to the exclusion of sparse-canopy biomes:

1. the limitations of spatial resolution, since an accurate model of ITC cover is hard to create with most publicly available satellite RGB datasets at 5–10 m/px. In practical usage, high ($<$ 1 m/px) or ultra-high ($\leq$ 0.1 m/px) resolution is required for ITC [33]. For reference, in the OAM-TCD holdout set used in this work (439 images, 25,705 individual tree crowns), downsampling from the images' native 0.1 m/px to a Ground Sample Distance (GSD) of 1 m (i.e. the real-world area corresponding to one pixel) results in 25.0% of tree crowns enclosing no complete pixel. At 2 m GSD, 57.0% of crowns enclose no complete pixel, and 22.6% have a bounding box narrower than a single pixel on at least one axis;
2. underrepresentation in high-quality annotated datasets. Conventional global tree cover maps such as Hansen et al. [34] have been shown to systematically underestimate tree cover in drylands [4] and are markedly less accurate outside closed-canopy forests [35]. Biomass and carbon products used for drylands [36–38] typically apply a single method to forest and dryland vegetation alike [2]. We speculate that this may also have led to a subtle vicious cycle: under-counting causes woody biomass in sparse-canopy biomes to appear too scarce to justify annotation effort, which in turn stymies further exploration of tree crown detection and instance segmentation in exactly the biomes where cover is most uncertain [5]. By way of illustration, in their study of 9.9 billion individual tree crowns in the Sahara/Sahel/Sudan region, Tucker et al. [2] found that "areas with scattered trees are often represented by zero values", attributing this to "the fact that previous models are rarely developed, trained and validated with plots of very sparse tree cover".

In short, datasets precede models (since existing models all rely on supervised learning), and as noted above, sparse-canopy environments have not historically been thought to contain enough trees to be worth the focus. Nevertheless, semi-arid ecosystems account for roughly half of the trend and 39–47% of the interannual variability in the global land carbon sink [39], and drive anomalous extremes such as the record 2011 sink [40]. Drylands as a whole (hyperarid, arid, semi-arid and dry sub-humid zones, using the United Nations Food and Agriculture Organization category definitions) cover $\sim$ 41.5% of the Earth's land surface [5] and are expected to expand from this figure by 10–23% by the year 2100 [41].

Things are changing, however. Recent datasets such as SelvaBox [3] & SelvaMask [23] have broadened geographic coverage of publicly available annotations, and the OAM-TCD dataset [33] in particular incorporates a variety of biomes across the globe, including urban, tundra, and savannah.

1.2. Reliance on manual mask annotations

Apart from limitations within sparse-canopy coverage, tree crown instance segmentation also remains understudied due to the lack of individual tree crown mask annotations in datasets [27]. Since crown polygons are slower and typically more expensive to annotate than boxes, box-supervised instance segmentation [42–44] offers a practical route to instance-level delineation without polygon labels. The recent rise of self-supervised backbones such as DINOv2 [45] and DINOv3 [46] has demonstrated that models pretrained on very large sets of diverse web imagery (1.7 billion images in the case of DINOv3-web) are capable of matching or even exceeding Earth-observation-specific foundation models (e.g. DINOv3-sat) on high-resolution RGB RS tasks, with the latter retaining an advantage chiefly in physically grounded regression such as canopy height estimation [46]. Apples-to-apples comparison across this literature remains difficult however, as papers variously report AP_{50}, $AP_{50:95}$, instance- or semantic-F1 at differing IoU thresholds against boxes and/or polygons. One of the contributions of this work is to present a direct comparison of the key SOTA models in the space.

1.3. Approach

In this work we propose a 3-stage model "LACE" built on a frozen DINOv3-web (ViT-L/16) [46] encoder, with a 3.2M-parameter CenterNet-style [47] detector head, and a novel lightweight expectation-maximisation module that leverages latent commonality of the ground truth bounding box annotations. In other words, we effectively ask the question: "can we leverage the vector space signature of the tree-ness within DINO latent feature vectors, based only on the commonality between rectangular bounding box examples, to isolate the trees/foreground from the background?" Our experiments across a range of biomes including temperate, tropical, xeric, and urban scenes, demonstrate that even with diverse tree species and highly variable background context, a latent commonality module is sufficient to infer and delineate objects of interest from rough rectangular annotations alone.

LACE achieves a mask AP_{50} of 0.647 ± 0.003 on the OAM-TCD holdout benchmark and a box F1 of 0.728 ± 0.003 on NeonTreeEvaluation (evaluated at IoU 0.4). The creators of NeonTreeEvaluation propose an IoU of 0.4 as a suitable threshold for tree crown detection tasks [17] based on comparative analysis of agreement and overlap between different human annotations of the same datasets, and we follow their convention on this dataset for direct comparison with their results.

By using only the hand-annotated bounding box annotations (which are sampled from 13 of the total 22 sites present in the test dataset), our model was able to match DeepForest's published results of 0.719 F1, evaluated at IoU 0.4. We also did not make use of a "canopy height model" (CHM) (a LiDAR derived height raster at 1 m spatial resolution), or any of the NeonTreeEvaluation dataset's hyperspectral imagery, since our focus is the underexplored and data-scarce sparse-canopy context. By demonstrating the efficacy of LACE on RGB data alone, we hope our contribution to the remote sensing community will have a broader application to data-scarce contexts.

In summary, the lack of publicly available tree crown-annotated datasets within sparse-canopy landscapes generates a need for box-supervised yet performant instance segmentation models suitable for diverse sparse-canopy contexts. Our contribution is deliberately orthogonal to dense-canopy tropical forest models, some of which are used as baselines for comparison in the Results section. In broad terms, most existing state-of-the-art models focus on improving the precision of delineation in dense environments. By contrast, our focus is on identifying tree crowns in sparse-canopy landscapes. Nevertheless, LACE is still able to demonstrate comparable performance in apples-to-apples comparisons across both dense- and sparse-canopy environments, with less training data and box supervision alone. By leveraging the rich DINOv3 features which enable cross-biome transferability, LACE is, to the best of our knowledge, the strongest performing box-supervised model in

the TCIS space, and is competitive with mask-supervised models trained on several times more images, as evidenced by results on the OAM-TCD benchmark.

In this work, we present:

- a) our model LACE, built on a DINOv3-web (ViT-L/16) backbone [46], with a trained CenterNet-style detector head [47] that identifies crown centres for detection based on the DINO output embeddings. We assess the performance of this approach against existing models on two publicly available datasets: Restor's OAM-TCD and DeepForest's NEON annotation dataset.
- b) within LACE, a novel box-to-mask module designed to extract the commonality in latent space, using an Expectation-Maximisation (EM) algorithm to isolate "treeness" within the bounding box. (Note that although it is this module from which we derive the name and could justifiably be called LACE by itself, it requires both encoder embeddings and also box delineations to distinguish instances from pure background, so we choose to apply the name to the model as a whole.) Unlike other existing implementations such as EM-Adapt [48], which amortises the E-step into a gradient-trained CNN with implicit cross-box sharing, we fit an explicit closed-form commonality mixture over frozen features. This approach exploits the similarities between signatures of composite embeddings within ground truth bounding boxes, thematically following previous works such as DiscoBox [49] and DDT [50]. In our implementation however, we align more closely with architectures such as STEGO [51] by utilising a World Model/JEPA [52] style approach, eschewing pixel-space calculations in favour of strictly latent space analysis. This approach is made possible by the capabilities of Meta's DINOv3 encoder foundation model [46], a Vision-transformer-based model trained on 1.7 Billion images using Self-supervised Learning, and we are indebted to their valuable contribution.

2. Related Work

The use of computer vision models to process aerial and satellite RGB imagery for the purposes of tree crown measurements has seen a number of advancements in recent years [32]. Two important benefits of RGB methods compared to other data modalities such as LiDAR or hyperspectral data-based methods are firstly the widespread availability of aerial RGB imagery, and secondly the high spatial resolution of that data, which for aerial platforms is typically centimetre-scale ($\leq$ 0.1 m/px, the ultra-high-resolution regime in which every method in Table 1 operates). Moreover, aerial RGB data can also be collected by UAVs/'drones' with relative ease. Consequently, in recent years there has been a focus in the RS community on RGB datasets which can directly leverage Computer Vision deep learning models [53] especially through the use of Convolutional Neural Networks (CNNs) [54,55] and Vision Transformers (ViT) [56].

2.1. Tree Crown Models

2.1.1. Convolutional Neural Networks

Modern CNN-based object detection models are comprised of three parts:

1. a feature extractor network ('backbone') which is a neural network that scans with convolutions to pick up on features in the image, with simpler features typically extracted earlier in the network's layers and more complex composite features requiring greater depth [57];
2. a 'neck', which can merge the detections across different layers of the backbone and provide access to the whole gamut of feature scale and complexity;
3. the Detector 'head(s)', which take the encoded features as inputs and produce the entire CNN's output, which can be, for example, bounding boxes around detected objects, polygons for a precise segmentation mask, or classification categories. Detector heads are generally either one-stage or two-stage detectors. Two-stage detectors first utilise a module to create many possible bounding boxes which are region proposals, before the second stage module extracts features from the proposals [58]. In

this way, the model can accurately detect within a large image by focusing on high-probability smaller regions.

One-stage detectors include YOLO ('You Only Look Once') series of models [59,60], RetinaNet [61] and EfficientDet [62]. Within TCD, the dominant two-stage detectors descend from the Region-based CNN (R-CNN) [63]: Fast R-CNN [64] shares convolutional features across proposals, Faster R-CNN [65] replaces external proposals with a learned Region Proposal Network (RPN), and Mask R-CNN [19] extends Faster R-CNN with a parallel per-instance mask head, which is what makes it the standard architecture for TCIS.

2.1.2. Star-convex

Star-convex representations offer a different route to separating touching crowns. Tong & Zhang [20] adapt StarDist [21] to tree crowns, replacing axis-aligned boxes with star-convex polygons so that NMS prunes overlapping proposals on crown geometry rather than box overlap, and report gains over Mask R-CNN and other SOTA models on dense, overlapping canopy in RGB imagery. MP-PolarMask [66], although not a TCIS-domain model, extends the star-convex representation with auxiliary polar centres to capture concave shapes, which could perhaps be suitable for non-star-convex crowns such as palm trees. FG-TreeSeg [67] takes a different route: it starts from a canopy semantic mask, then applies Cellpose-SAM, a cell-segmentation model that predicts, for every canopy pixel, a flow vector pointing towards its crown centre. By following these vectors, pixels can be grouped by the centre they converge on, so touching crowns are split into instances without any instance annotations.

2.1.3. DeepForest

DeepForest [16] is a significant baseline model in Tree Crown Detection (TCD). DeepForest is a RetinaNet CNN on a ResNet50 backbone and Feature Pyramid Network (FPN) trained on over 30 million LiDAR-prompted algorithmically generated crowns from 22 National Ecological Observatory Network (NEON) sites and further fine-tuned using 10,000 hand-annotated crowns.

2.1.4. Pretrained Foundation Models

Detectron2 [68] is Facebook AI Research's object detection and segmentation library rather than a model in itself: it supplies reference implementations of Mask R-CNN, RetinaNet and Faster R-CNN together with the training, inference and evaluation harness used to fine-tune them. Several of the tree crown baselines in this work are built on it, including Detectree2 [18] and the OAM-TCD Mask R-CNN released by Restor [33]. Its model-zoo checkpoints, from which those baselines were fine-tuned, are trained on the 118k-image COCO [69] `train2017` split from backbones pretrained on the 1.28M-image ImageNet-1k [70] classification set, which is what makes them a generic, domain-agnostic starting point.

Meta's Segment Anything Model (SAM) [25], now in its third iteration [71], is a promptable class-agnostic segmentation foundation model: given a point, box or mask prompt it emits an instance mask, which makes it a natural box-to-mask module for detectors that emit only rectangles, and it is used in that role by the SelvaMask model, one of baselines compared here.

Meta's latest version of the Self-Distillation with No Labels (DINO) family [45,72] is DINOv3 [46]: a SOTA encoder trained with fully Self-Supervised Learning (SSL) on 1.7 billion images pulled from various online sources. Utilising SSL facilitates training regimes on orders of magnitude more image data than supervised learning, since weakly and fully supervised methods typically require high-quality labels and/or metadata for the training images [73–75]. Highly context-agnostic, DINOv3-web is an example of a frozen SSL backbone that can serve as a "universal visual encoder" [46] capable of generating rich embedding vectors that capture the semantic information from the raw pixel patches in context. Incorporating both a global image-level objective and a local iBOT-style [76] patch-level

latent reconstruction objective in the loss function results in a model that excels at encoding global and local features. In this work, we use the DINOv3-web ViT-L distilled model, with a native 16 px patch.

2.1.5. Selvabox

SelvaBox [3] is a UAV-captured RGB dataset of tropical dense-canopy forest in Panama, Brazil, and Ecuador, composed of 83,000 ITC manual bounding box annotations. Its authors publish a DINO (DETR) [24] Swin-L [77] based tree detection model (also referred to hereafter as 'SelvaBox' following its authors' own convention) trained on NeonTreeEvaluation, OAM-TCD, and another dataset QuebecTrees, which achieves a best result of box $mAP_{50:95}$ of 44.29 ± 0.33 on OAM-TCD holdout set, improving upon DeepForest (39.00 ± 0.21) and a Faster R-CNN ResNet50 (38.34 ± 0.26) finetuned on the same data. A second related dataset, SelvaMask [23], provides ITC instance segmentation masks and assesses the SelvaBox detection model in combination with both frozen and fine-tuned SAM 3 decoder modules on the OAM-TCD dataset for the TCIS task. The authors compare results on a number of benchmark datasets including OAM-TCD alongside Detectree2 and DeepForest with a similar frozen SAM 3 mask prediction module. SelvaMask's focus is increased precision of tree crown boundaries in dense canopy, whereas our focus is on sparse-canopy detection accuracy. They note that their model is specialised "to dense, interlocking canopies, slightly reducing its transferability to temperate forests".

Leveraging SAM as an out-of-the-box instance segmenter for remote sensing has been explored in RSPrompter [26], which overcomes SAM's reliance on prompts such as points/boxes/masks by training a prompter module that learns the intermediate layer features of the SAM encoder on labelled training images, and then using those embeddings as input to SAM's mask decoder in order to produce category-labelled instance masks.

Table 1. Representative individual tree crown methods, grouped by the annotation type used for training. The table maps the design space rather than surveying it exhaustively: a method is included if it emits per-tree boxes or instance masks from ultra-high-resolution airborne imagery ($\leq$ 10 cm GSD, the regime in which ITC delineation is feasible), and rows were chosen so that every supervision regime and every biome column is represented by at least one published method. Methods used as baselines in this work are marked *. *Output*: boxes or instance masks. *Extra net(s)*: pretrained networks required at inference beyond the method's own backbone. Biome coverage refers to evaluation.

				GSD	Biome		
Method	Output	Input	Extra net(s)	(cm)	Tr	Te	Sv
Trained on crown polygons							
Mask R-CNN (OAM-TCD)* [33]	Mask	RGB	–	10	○	○	○
Detectree2* [18]	Mask	RGB	–	10	●	×	×
Mask R-CNN / DETR [78]	Mask	MS+LiD	–	5	×	●	×
CrownViM [79]	Mask	RGB	–	10	○	○	○
BalSAM [27]	Mask	RGB+DSM	SAM	<5	●	●	×
SelvaMask* [23]	Mask	RGB	SAM 3	1–4	●	●	○
Trained on bounding boxes							
DeepForest* [16]	Box	RGB	–	10	○	●	●
SelvaBox* [3]	Box	RGB	–	1–5	●	●	○
LACE (ours)	**Mask**	**RGB**	**–**	**10**	○	●	●
No instance annotations							
SAM 2 + LiDAR prompts [80]	Mask	RGB+LiD	SAM 2	10	×	●	×
FG-TreeSeg‡ [67]	Mask	RGB	SegFormer, Cellpose-SAM	≤ 10	×	●	×

Biome columns are groups of WWF terrestrial biomes, the classification used throughout this paper: Tr, tropical forest (WWF 1–3 and 14); Te, temperate and boreal forest (WWF 4–6 and 11); Sv, savannah/dryland, i.e. the sparse-canopy group (WWF 7–10, 12 and 13) from which the sparse-canopy slice of Table 6 is drawn. ● evaluated and reported separately; ○ present in an aggregate benchmark (e.g. OAM-TCD), not reported as a stratum; × not evaluated. LiD, LiDAR; MS, multispectral; DSM, digital surface model. ‡Trained on semantic canopy labels only; instances are recovered without instance annotations.

2.1.6. Box-supervised mask inference

There is a growing class of box-supervised architectures which, as the name suggests, utilize only box annotations during the training phase but are able to produce finegrained mask level delineations. This is the category in which LACE fits most directly, although to the best of our knowledge, the application of these architectures to TCIS has not been explored in detail. BoxInst [42] established the modern recipe by replacing the pixel-wise mask loss of a CondInst-style network with two terms derived from the box alone: a projection loss requiring the horizontal and vertical projections of the predicted mask to match the box edges, and a pairwise affinity loss encouraging neighbouring pixels of similar colour to share a label, reaching 33.2 mask AP on the COCO benchmark [69] (the standard 80-class natural-image instance segmentation benchmark on which these methods are ranked) against 39.1 for its fully mask-supervised counterpart. DiscoBox [49] supervises the student network with a structured energy teacher whose pairwise and cross-image potentials refine masks within boxes while jointly yielding semantic correspondences between instances of the same class. BoxSnake [81] instead predicts polygon vertices directly and supervises them with a box-edge matching term plus local and global pairwise consistency terms. Box2Mask [43], used as a baseline in our experiments, casts the mask inside each box as a level-set function evolved implicitly by the network, with an energy defined over both the input image and its deep features and a pixel-affinity local consistency module; with a Swin-L [77] backbone it achieves 42.4 mask AP on COCO, which its authors describe as on par with recent fully mask-supervised methods.

3. Materials and Methods

3.1. Datasets

Restor OAM-TCD

The OAM-TCD dataset [33], introduced by Veitch-Michaelis et al. (ETH Zurich, Restor and collaborators), uses OpenAerialMap (OAM) [82] RGB imagery and provides 5072 2048 $\times$ 2048 px images at 0.1 m/px resolution with associated human-labelled instance masks for over 280k individual and 56k groups of trees (groups are hereafter referred to as "canopy"). In our work, the OAM-TCD dataset has been an invaluable benchmark to assess the comparative performance of our LACE model in an apples-to-apples comparison against other SOTA models, due to the inclusion of not only bounding boxes but also precise polygon masks, as well as its broad range of ecological biomes to capture the diversity and morphology of trees in different terrestrial biomes including both urban and natural environments. This multi-region, multi-biome distribution of annotated aerial RGB data allows for a much more diverse training set which aids cross-biome transfer, and also contains examples of the savannah & dryland landscapes so scarce in existing annotated datasets.

Of the 5072 images in the OAM-TCD dataset, 439 are reserved as a holdout benchmark test set, allowing for a global multi-biome comparison between models. The 439-image test set contains 25,705 ITC annotations, and 366 images within the set contain ≥ 1 crown, i.e., 73 of the images have 0 ITCs. The same test set also contains 6,782 canopy annotations, which are groups of trees in a wide range of real-world sizes. 364 of 439 tiles contain ≥ 1 canopy region. During creation of the dataset, human annotators were told to use the canopy class when it was not possible to reliably delineate individual trees, including when it was "not obvious whether a tree was an individual or multiple". 342 images contain at least one of each class and 51 tiles are fully empty. The densest tile holds 422 individual crowns, and the median tile holds 29.

NeonTreeEvaluation

The U.S. National Science Foundation's National Ecological Observatory Network (NEON) publicly available aerial imagery dataset is a cornerstone of RS research, comprising 22 separate locations across the continental United States and including a variety of different biomes. The DeepForest model released alongside the NeonTreeEvaluation benchmark [17] was pretrained on all 22 sites, using an unsupervised LiDAR-based algorithm [83] to generate millions of moderate-quality annotations, and was then fine-tuned on a further 10,000 manual annotations of RGB imagery from six sites. The quality and scale of the NEON dataset as well as its fundamental role in the development of the DeepForest model was a key reason to build our pipeline to natively support the NEON resolution of 0.1 m/px.

Data preprocessing: We pinned NeonTreeEvaluation @1.8.0 and built the evaluation GT by intersecting annotation XMLs with RGB tiles. This resulted in 194 scored tiles / 6,634 boxes / 22 sites. We downloaded the hand-annotated RGB training tiles, which are spatially disjoint from the evaluation tiles, and cropped 18 kept tiles (13 sites, 23,424 source boxes) into 2,063 non-overlapping 400 px patches (26,800 boxes after clipping to patch bounds), restricted to genuinely-annotated regions only and capping empty negatives at 0.3$\times$ positives per tile. Two tiles were dropped due to low annotation coverage (SJER 41.8% / TOOL 0% coverage). The train/val split is patch-level stratified (every 8th patch of each tile, 1,807 train / 256 val), so all sites appear in training; val is used only as the early-stopping signal and the reported metric is the fully disjoint 194-tile benchmark.

Both are RGB-only (we did not use the dataset's LiDAR/CHM/HSI data at any time), at native 0.1 m/px, and the 400 px patch geometry matches the 400 px evaluation tiles so training and test see the same scale. Features are then extracted by padding each patch to 512 px and encoding it to a 32 $\times$ 32 DINOv3 grid per pass (64 $\times$ 64 after interlacing), taking layers 21–24 (4096 dimensions, against the single final layer used on OAM-TCD; Section 3.2.1).

3.2. LACE

Our method uses a frozen self-supervised encoder to produce instance masks for individual tree crowns using bounding-box supervision only. It has three stages: a dense feature grid obtained by interlacing four offset passes of a frozen DINOv3-web ViT (Section 3.2.1); an anchor-free detector trained on bounding boxes (Section 3.2.2); and a training-free box-to-mask module that converts each predicted box into a mask by foreground/background discrimination in a whitened feature space (Section 3.2.3). Predictions are then ranked by combining the detector score with the mask module's own posterior (Section 3.2.5), which adds no parameter and leaves the masks untouched. No crown polygon is read at any stage of encoding, detection or mask fitting. The only crown mask labels used anywhere in the pipeline are the validation images on which the inference settings are selected (Section 3.2.4). Figure 2 follows a holdout example through the pipeline and expands the box-to-mask module into its fit and inference steps.

3.2.1. Dense features by phase interlacing

A ViT with patch size p emits one embedding per $p \times p$ patch, giving a feature grid p times coarser than the image. For crowns whose diameter is a small multiple of p this is the binding resolution constraint i.e. the mask raster resolution is effectively limited to the feature grid 'resolution'. A method like bilinear upsampling increases the sampling rate but introduces no new measurements.

Rather than relying on alternative approaches such as upsampling which increases the sample resolution by synthesising visual information, we instead evaluate the frozen encoder four times per tile, shifting the input by $(\delta_y, \delta_x) \in \{0, p/2\}^2$, and interlace the four grids, which we found produced higher performance than upsampling (Section 4.1). With $p = 16$ this yields a real 8 px lattice. Writing $F^{(\delta_y,\delta_x)}$ for the grid from one offset, the assembled feature tensor $\widetilde{F}$ is

$$\widetilde{F}\Big[:,\ 2i + \tfrac{\delta_y}{8},\ 2j + \tfrac{\delta_x}{8}\Big] = F^{(\delta_y,\delta_x)}[:, i, j], \qquad i, j \in \{0, \ldots, 127\} \tag{1}$$

so that cell (Y, X) of $\widetilde{F}$ carries the embedding of the patch centred at pixel $(8X + 8,\ 8Y + 8)$. Each cell is therefore a genuine encoder output at its own location with a virtual 4 px border included, rather than an interpolation of its neighbours, and the union of the four phases' patch centres is exactly the 8 px lattice.

Encoding uses DINOv3 ViT-L/16 with weights frozen throughout. On OAM-TCD we take final-layer (layer 24) activations (1024 dimensions; on NEON, layers 21–24, 4096 dimensions), giving $\widetilde{F} \in \mathbb{R}^{1024\times256\times256}$ for a 2048 px tile. Registration is verified independently in three ways: a synthetic blob is localised to within one cell across 200 positions; the cells of $\widetilde{F}$ belonging to the unshifted phase reproduce a plain single-pass extraction to floating-point precision, confirming the indexing of Equation 1; and the cells contributed by the three shifted passes agree with a bilinear upsampling of the unshifted grid only to a cosine similarity of ≈ 0.955. Were the shifted passes redundant, interpolation would reproduce them almost exactly, so the departure from unity is the evidence that they carry measurements which upsampling cannot synthesise.

3.2.2. Detection

Detection uses an anchor-free centre-based head operating directly on $\widetilde{F}$ at stride 8. A 1×1 stem projects 1024 channels to width 256, followed by five 3×3 convolutional blocks with group normalisation and ReLU (3.2M parameters in total; 4.0M with the 4096-dimensional NEON input), and two 1×1 output heads: a single-channel centre heatmap and a four-channel regression map carrying a local sub-cell offset and log box size. This is based on the CenterNet design [47]. The heatmap bias is initialised to -2.19 so the initial centre probability is ≈ 0.1, and the size biases are initialised to a 20 px crown at stride 8.

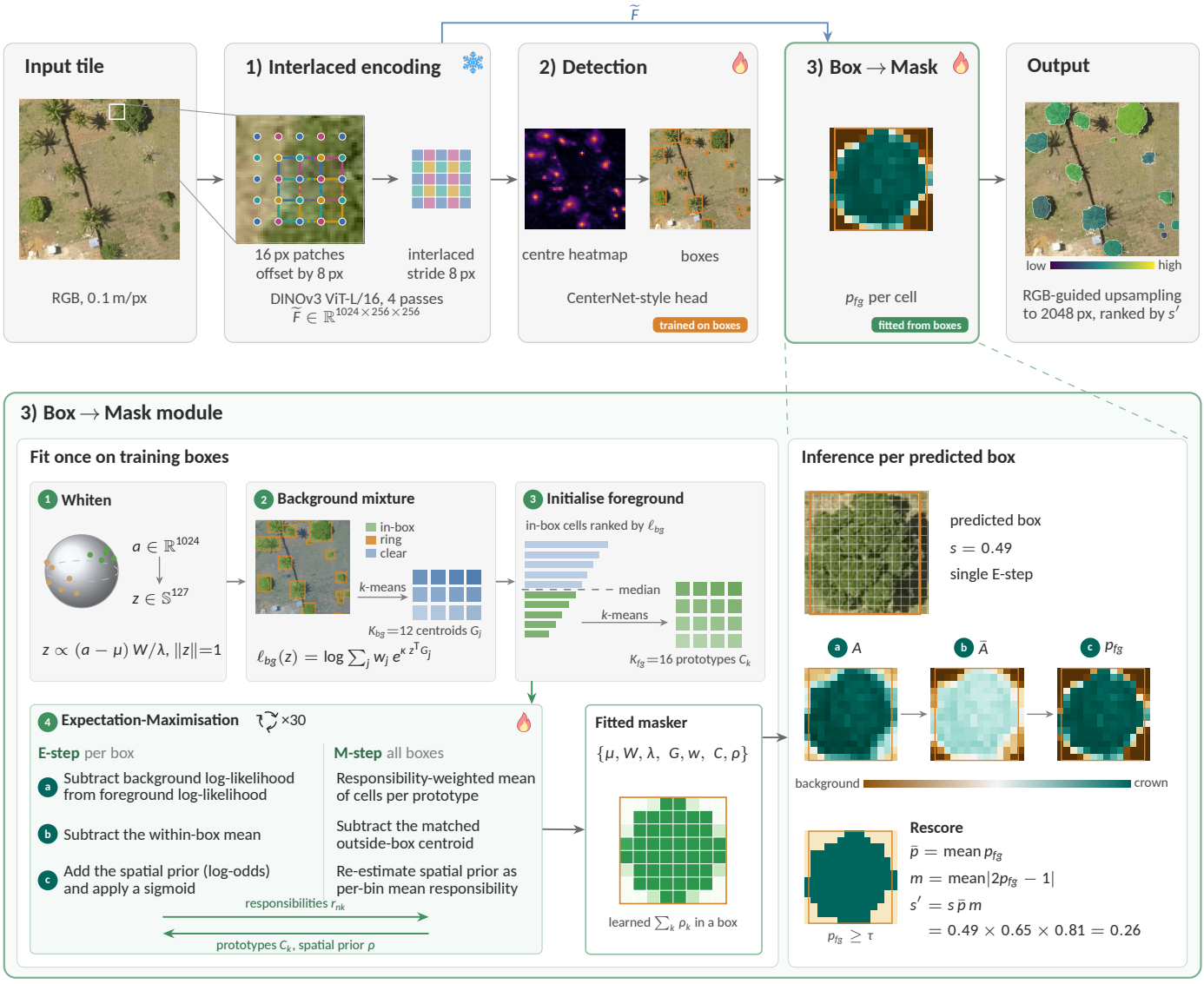


Figure 2. LACE architecture on an OAM-TCD holdout crop. Top: (1) a frozen DINOv3 encoder is run at four 8 px offsets and interlaced into a stride-8 feature grid, (2) a detection head trained on boxes predicts crown boxes, and (3) each box is converted into a mask, upsampled to the native tile with an RGB-guided filter, and ranked by its rescored confidence s'. Bottom: the box-to-mask module is fitted once on training boxes by (1) whitening the features, (2) fitting a background mixture, (3) seeding foreground prototypes and (4) learning the prototypes and spatial prior by EM. At inference each predicted box receives a single E-step and is rescored.

Training minimises a focal loss on the centre heatmap plus smooth-L_1 losses on offset and size at annotated centres, and the size term is weighted by 0.1. Because the interlaced grid is already at the target stride, the head contains no internal upsampling; the half-cell difference between the assembled cell centre $8X + 8$ and the target cell centre $8X + 4$ is a uniform translation absorbed by the offset head, whose bias is initialised accordingly. Centre targets are encoded as CenterNet Gaussians whose radius follows the standard IoU criterion of CornerNet [84] at an overlap of 0.7, clamped never to fall below one feature cell so that the smallest crowns retain a non-degenerate target. Boxes are decoded by local-maximum selection on the heatmap, retaining candidates scoring above 0.05 up to a maximum of 600 per tile. Average precision is computed over the full ranked list; where

precision, recall and F1 are reported at a single operating point which is a detector score of 0.4.

3.2.3. Training-free box-to-mask conversion

Given a detection box, the task remains to decide which of its cells belong to the crown. To be completely explicit about the problem, a valid box will almost always contain pixels of the tree itself, and also of the ground or surrounding objects as well, purely as a consequence of the geometry of rectangular boxes relative to the typical shape of trees when viewed from the air. We treat this as foreground/background discrimination in a whitened feature space, with the model fitted once on training tiles (using only boxes) and then held fixed.

The module has four key components: a whitened feature space; a frozen background mixture; a set of foreground prototypes; and a spatial prior over relative position within the box. The last two are learned by an alternating EM-based fit, while the first two stay fixed.

Feature space

LACE's core idea is to treat the rich DINOv3-web embeddings as sufficiently descriptive representations of each patch to build a high-performing tree-crown instance segmentation model. We begin with a, a given cell's raw unmodified embedding, and PCA-whiten it: a is centred, projected onto the leading $d = 128$ principal directions of the training-cell distribution (pooled across all tiles and boxes) and rescaled so that each retained component has unit variance. A final L_2 normalisation then places the result on the unit sphere, giving the unit vector z.

$$\tilde{z} = \frac{(a - \mu)\, W}{\lambda}, \qquad z = \frac{\tilde{z}}{\|\tilde{z}\|} \tag{2}$$

Here $a \in \mathbb{R}^{1024}$ is one cell of $\widetilde{F}$; $\mu \in \mathbb{R}^{1024}$ is the cell mean, and $W \in \mathbb{R}^{1024\times d}$ is the whitening matrix, whose columns are the retained principal directions. $\lambda \in \mathbb{R}^{d}$ is the vector of per-component standard deviations, which are all estimated once on training cells and then frozen. This whitening/spherising equalises the variance of the retained directions so that no direction dominates by scale alone. The final normalisation then places every cell on the unit sphere, making all subsequent comparisons simple cosine similarities and allowing both foreground and background to be modelled by von Mises–Fisher components [85] with a shared concentration κ.

Background mixture

Cells are partitioned into three categories: in-box cells, clear background cells, and a special category of cells immediately outside each box, in a ring. The ring cells share illumination, phenology and local context (e.g. ground shadow) with the crown and are therefore suitable hard negatives. A background mixture of K_{bg} components $\{(w_j, G_j)\}_{j=1}^{K_{bg}}$ is fitted to the union of background and ring cells by spherical k-means and *frozen*, giving the background log-likelihood ℓ_{bg} as follows:

$$\ell_{bg}(z) = \log \textstyle\sum_j w_j \exp\!\big(\kappa\, z^\top G_j\big) \tag{3}$$

Each background centroid G_j is a unit vector on the same sphere as the cells, the mixture weights w_j sum to one, and $z^\top G_j$ is therefore just a cosine similarity. The log-sum-exp acts as a smooth maximum over the components, so $\ell_{bg}(z)$ is a single scalar per cell reading as how well that cell matches its best-matching background component. It is neither a

threshold nor an objective, but the evidence term that Equation (4) scores the foreground against.[1]

Foreground mixture

Foreground and background take the same von Mises–Fisher form with a shared concentration κ, and differ in one respect only: the background weights w_j are global and frozen, whereas the foreground weights are supplied per cell by the spatial prior and re-estimated at every M-step. Foreground is represented by K_{fg} unit prototypes $\{C_k\}_{k=1}^{K_{fg}}$, the exact counterpart of the background centroids G_j, initialised by spherical k-means over in-box cells restricted to those atypical of the background, i.e. cells which have a value of ℓ_{bg} below the median ℓ_{bg} of all in-box cells. This is merely a heuristic to kick-start the fitting. Without this restriction, the initialisation is seeded from in-box background elements such as soil and roads, which are numerous and not the intended crown foreground.

Spatial prior

We observe that a crown typically occupies a roughly central and roughly circular region of a reasonably tight GT box. We encode this as a spatial prior ρ over the crown's *relative* position within the box, to push the model towards producing reasonable tree crown-like prediction masks. In particular, we aim to increase LACE's robustness to dense-canopy and lush undergrowth situations, where the box-to-mask module can unintentionally produce masks that are not tree crown-like. The box is divided into an $N \times N$ grid of bins in normalised coordinates $(u, v) \in [0, 1]^2$. Because those coordinates are a *fraction* of the box, ρ is by construction invariant to crown size, tile size and ground sample distance. Priors are maintained separately for three crown-size bands, whose edges are the terciles of the training box-size distribution, since at a fixed feature stride the number of cells spanning a crown depends on its size. Taken together, ρ therefore carries one value for each (size band, prototype, u bin, v bin) combination: it is a joint prior over where in the box a cell sits and which prototype is expected there. We write $\hat{\rho}_k = \rho_k / \sum_{k'} \rho_{k'}$ for the prior at a bin normalised across prototypes, so that $\hat{\rho}$ is a distribution over the K_{fg} prototypes at that bin while the unnormalised total $\sum_k \rho_k$ remains the probability that the bin is crown at all. Equations (4) and (6) draw on these two quantities separately. ρ is initialised uniform, so all prototypes begin with identical spatial profiles and differentiate only through Equation (10).

Two constraints are applied to ρ at every M-step. The first pins its mean per-bin foreground mass, summed over prototypes and averaged over the N^2 bins, to $\pi/4 \approx 0.785$, the area fraction of an ellipse inscribed in its bounding box and hence independent of box aspect ratio. The second caps any single bin at 0.95, so that no relative position is ever treated as certainly foreground. Pinning the mass to a constant derived from geometry rather than fitted is what keeps the prior label-free, and it means the fit decides only *where* the foreground mass sits, never *how much* of it there is: without the constraint ρ inflates towards unity, stops discriminating position, and the α term of Equation (6) degenerates into an additive constant.[2]

[1] Because G_j, w_j and κ are all fixed once the mixture is frozen, and z depends only on the frozen whitening of Equation (2), every input to Equation (3) is determined before fitting begins: ℓ_{bg} is therefore computed once per tile, cached, and reused unchanged by every iteration. Note also that the standard von Mises–Fisher normalising constant $C_d(\kappa)$ is omitted here and from the foreground term of Equation (4) alike; sharing a single concentration parameter κ between the two mixtures makes it identical on both sides, so it cancels in the difference and never has to be evaluated in $d = 128$ dimensions. Equations (3), (4) and (7) are written in the conventional mixture form $\sum_j w_j \exp(\cdot)$ for readability; all three are evaluated in log-sum-exp form for numerical stability.

[2] The constraints are applied as a single rescale followed by the cap, so the cap partly undoes the rescaling: on the deployed fit this leaves leaving a realised mean per-bin mass of 0.690 against the nominal 0.785.

Fitting

Once we have these objects (the whitening (μ, W, λ), the frozen background mixture $\{(w_j, G_j)\}$, the foreground prototypes $\{C_k\}$, and the spatial prior ρ), we can run the EM fitting algorithm, which alternates a per-box E-step with a global M-step, set out in full in Algorithm 1. Only $\{C_k\}$ and ρ are learned here, the rest are estimated once and held fixed.

E-step. For the cells of one box, with ρ_k the spatial prior evaluated at each cell's bin for prototype k and $\sum_k \rho_k$ the total foreground prior at that bin, the posterior follows in three steps: an appearance *log-ratio* (4), its within-box *recentring* (5), and the *posterior* itself (6).

$$A(z) \;=\; \log \textstyle\sum_k \hat{\rho}_k \exp\!\big(\kappa\, z^\top C_k\big) \;-\; \ell_{bg}(z) \tag{4}$$

$$\bar{A}(z) \;=\; A(z) - \mathrm{mean}_{z' \in \mathrm{box}}\, A(z') \tag{5}$$

$$p_{fg}(z) \;=\; \sigma\big(\bar{A}(z) \;+\; \alpha\, \mathrm{logit}\big(\textstyle\sum_k \rho_k\big)\big) \tag{6}$$

Equation (4) is a log-likelihood ratio. Its first term is the foreground log-likelihood, built from the prototypes C_k exactly as Equation (3) is built from the background centroids G_j; subtracting ℓ_{bg} gives $A(z) = \log\big(p(z \mid \text{crown}) / p(z \mid \text{background})\big)$. A cell must therefore look crown-like *and* unlike background to score highly. This is the first of two filters that in-box soil/ground has to pass, and it removes the soil/ground that resembles the background mixture outright.

Equation (5) is the second filter, and the step that makes the model work on frozen ViT features. Problematically, attention over the surrounding tile inflates the absolute foreground score of *every* cell inside a box, so A is confounded by context and its absolute level is not sufficiently informative: e.g. soil surrounded by canopy can still score positively, and therefore survive the first filter. However, recentring isolates A *within* a box, asking not whether a cell is crown-like but whether it is *more* crown-like than the rest of its own box. The behaviour that follows is essentially precisely what we wanted to produce: a heterogeneous box containing crown and soil has a wide spread of $\bar{A}$ and is carved appropriately, whereas a box in uniformly closed canopy has a narrow spread and stays closer to filled, which is generally correct.

Equation (6), the posterior, adds the prior mass term, and it is worth noting that the spatial prior enters the two equations in different roles. In Equation (4) it appears only as the normalised weight $\hat{\rho}_k$, which sums to one over k and therefore selects *which* prototype applies at a given relative position while discarding *how much* total foreground prior there is. That discarded total re-enters on its own in Equation (6), where the prior weight α scales it. The total foreground prior thus enters the posterior only through the α term of Equation (6). Equation (6) gives the probability that a cell is crown, but the M-step also needs to know *which* prototype accounts for it. Alongside p_{fg}, the E-step therefore returns a responsibility r_{nk} for each cell n of the box (with whitened embedding z_n) and each prototype k:

$$r_{nk} \;=\; p_{fg}(z_n)\; \frac{\rho_k \exp\!\big(\kappa\, z_n^\top C_k\big)}{\sum_{k'} \rho_{k'} \exp\!\big(\kappa\, z_n^\top C_{k'}\big)} \tag{7}$$

which together distribute each cell's foreground probability over the K_{fg} prototypes in proportion to the weighted components of Equation (4), so that $\sum_k r_{nk} = p_{fg}(z_n)$. In other words, r_{nk} is the probability that cell n is crown and is explained by prototype k, which the M-step uses to update the prototypes and spatial prior. (Note that Equation (7) is written with ρ_k rather than $\hat{\rho}_k$ because the normalisation appears in both numerator and denominator and therefore cancels.) Equations (4)–(7) are evaluated for all the cells of one box at a time and the results accumulated across all training boxes.

M-step. The prototypes are updated in two steps. The first is the ordinary generative pull: prototype k is drawn towards the responsibility-weighted mean of all cells over all boxes,

$$P_k \;=\; \frac{\sum_n r_{nk}\, z_n}{\big\|\sum_n r_{nk}\, z_n\big\|} \tag{8}$$

which expresses the idea "recurs across boxes". The second move repels the prototype from the outside-box signature it most resembles. The outside-box cells (clear background and ring) are softmax-assigned to the current prototypes at the same concentration κ; the resulting per-prototype negative centroid $\bar{N}_k$ is the normalised weighted mean of the cells assigned to prototype k, and the prototype becomes

$$C_k \;=\; \frac{P_k - \beta\,\bar{N}_k}{\|P_k - \beta\,\bar{N}_k\|} \tag{9}$$

A soil-like prototype's nearest negatives are themselves soil, so the subtraction tears it away from soil: commonality is defined as agreement-across-boxes *minus* outside-box signature, not merely whatever fills the box. The contrastive weight β controls the strength of that repulsion; $\beta = 0$ recovers the purely generative update of Equation (8) (compared in Section 4.1), and we use $\beta = 0.5$ throughout.

The spatial prior is re-estimated from the same responsibilities. Writing $b(n)$ for the bin of cell n and N_b for the number of cells falling in bin b,

$$\tilde{\rho}_k(b) \;=\; \frac{1}{N_b} \sum_{n\,:\,b(n)=b} r_{nk} \tag{10}$$

accumulated separately per size band, after which $\tilde{\rho}$ is rescaled to the $\pi/4$ mean per-bin mass and capped at 0.95 per bin as described above to give ρ. Finally, after a burn-in of five iterations, any prototype whose share of the total foreground responsibility, $\sum_{n,b} r_{nk} / \sum_{n,b,k'} r_{nk'}$, falls below $0.25/K_{fg}$ (a quarter of the uniform share $1/K_{fg}$) is deleted and K_{fg} decreases accordingly; pruning is skipped once $K_{fg} \le 2$.

Fitting is deterministic given the seed. The procedure is not gradient-based, reads no crown polygons, and produces a compact set of parameters $\{\mu, W, \lambda, C, G, w, \rho\}$.

Algorithm 1 Box-only masker fit. Inputs: training tiles with GT boxes.

Whitening. Estimate μ, W, λ on training cells; map every cell by Equation (2).
Background. Partition cells into in-box, ring and clear background by box geometry; fit $\{(w_j, G_j)\}$ on ring $\cup$ clear by spherical k-means; **freeze**. Precompute ℓ_{bg} once (Equation (3)); it never changes.
Initialise C**.** Spherical k-means over in-box cells with ℓ_{bg} below its median, giving K_{fg} prototypes.
Initialise ρ**.** Uniform, at $\pi/4$ / K_{fg} in every bin of every size band.
for $t = 1$ **to** $T = 30$ **do**
    for each training box **do** ▷ E-step, per box
        Evaluate A, $\bar{A}$, p_{fg}, r_{nk} by Equations (4)–(7).
        Accumulate $\sum_n r_{nk} z_n$ and the per-bin sums of r_{nk}.
    end for
    $P_k \leftarrow$ Equation (8); $C_k \leftarrow$ Equation (9). ▷ M-step, global
    $\rho \leftarrow$ Equation (10), then rescale to $\pi/4$ and cap at 0.95.
    if $t > 5$ **then** delete prototypes whose responsibility share is below $0.25/K_{fg}$ (a quarter of uniform), keeping at least two.
    end if
end for
return $\{\mu, W, \lambda, C, G, w, \rho\}$.

This loop has two important properties. First, the background mixture is frozen after step 2, so ℓ_{bg} is a constant of the fit rather than a quantity being jointly estimated. Second, strictly speaking, the procedure is EM-style rather than EM proper: the within-box recentring of Equation (5), the contrastive term of Equation (9) and the projection of ρ onto the mass and cap constraints each break the correspondence with a joint likelihood,

so no objective is monotonically ascended and there is no convergence test: T is fixed at 30 iterations.

3.2.4. Inference

For each predicted box, cells within the box are projected by Equation (2), the matching size band is selected, and posterior p_{fg} is evaluated by Equations (4)–(6). The posterior lives on the 8 px feature grid, so we upsample it to the native 2048 px tile with a guided filter [86] that uses the RGB tile as its guide, so that mask boundaries follow edges visible in the image rather than the feature grid (radius 12 px, i.e. 1.5 feature cells; $\epsilon = 10^{-4}$). Pixels above a mask threshold τ form the predicted tree crown mask, which is intersected with the box, so a mask is contained in its box by construction rather than by penalty.

Three scalars are exposed at inference, and together with the guided filter's radius and ϵ they are the only quantities in the pipeline selected against mask labels. The prior weight α scales the box-derived spatial term in Equation (6): values ≤ 1 relax reliance on a *predicted* box, which is not expected to always be perfectly precise, so that the appearance term dominates. (Because the spatial prior is defined in normalised box coordinates, α is invariant to crown size and GSD. What it tracks is the precision of the boxes it is applied to). The concentration κ is raised from the value of 10 used during fitting to 16 at inference, in Equations (3) and (4) alike; since $\bar{A}$ is zero-mean within each box, this scales the spread of the appearance term and thereby sharpens it. 10 was fixed a priori, and 16 is simply the best of $\{10, 13, 16, 20\}$ on validation, a range over which mask AP_{50} varies by under 0.005. The robust finding is directional: any $\kappa > 10$ outperforms $\kappa = 10$ at every α.

The mask threshold τ is the posterior cut that defines the mask, lowered from an initial 0.5 so that small crowns under-covered by imprecise predicted boxes are recovered. All three, and the radius and ϵ of the guided filter, are selected on the 108 validation images held out from the 900 training images (the detector trains on the remaining 792) under a rule fixed in advance, and none requires refitting. We therefore describe the method as trained without mask supervision but broadly calibrated for tree crown mask generation on a small held-out set of mask labels, rather than as entirely label-free. We take these to be reasonable 'tree crown' hyperparameters and do not attempt to optimise them for use on other datasets such as NeonTreeEvaluation.

3.2.5. Confidence from the mask posterior

Now that we have the mask generated, we want to update the confidence score of each tree crown rather than just retain the score provided by the detector head, which is $s = \sigma(h)$, the sigmoid of the centre heatmap logit h at the decoded peak of Section 3.2.2. This value correlates to whether a crown centre is present at a pixel within the box, but does not necessarily correctly score the resulting mask (from Section 3.2.3). However, we have already calculated the posterior p_{fg} of Equation (6) for every cell of a box. To generate a confidence score we can reduce the posterior over the $|B|$ cells of a box B to two terms which we can use as multipliers on the $\sigma(h)$ score to produce a new score $s'(B)$ which reflects the updated confidence based on LACE's experience of the training boxes it has seen.

$$\bar{p}(B) = \frac{1}{|B|} \sum_{n \in B} p_{fg}(z_n) \tag{11}$$

$$m(B) = \frac{1}{|B|} \sum_{n \in B} \left| 2\, p_{fg}(z_n) - 1 \right| \tag{12}$$

Because the recentring of Equation (5) sets the mean of $\bar{A}$ to zero within every box, both statistics read the *shape* of the within-box posterior rather than its level. For $\bar{p}$ this does not force the value $\frac{1}{2}$, since the mean of a sigmoid is not the sigmoid of the mean: writing $\sigma(x) = \frac{1}{2} + g(x)$ with g odd and saturating, $\bar{p}$ is $\frac{1}{2}$ plus the mean of $g(\bar{A})$ over

the box, up to the offset contributed by the spatial term of Equation (6), and that mean vanishes only when $\bar{A}$ is symmetric. $\bar{p}$ therefore records the *asymmetry* of the evidence. Where a minority of cells sits far above the box mean, the saturation of σ compresses their contribution and the mild majority below the mean dominates, giving $\bar{p} < \frac{1}{2}$; a box filled by crown apart from a few much darker cells gives $\bar{p} > \frac{1}{2}$. Read this way $\bar{p}$ is a soft estimate of the fraction of its box that the crown occupies. The decisiveness m folds each cell to its distance from $\frac{1}{2}$ before averaging, and so records the *spread* of the same distribution: $m = 1$ when every cell is confidently inside or outside, $m = 0$ when every cell is a coin flip. Folding before averaging is what makes it a measure of spread rather than of level, since a box split half confidently-crown and half confidently-background gives $m \approx 1$ but $\bar{p} = \frac{1}{2}$. The two are near-orthogonal descriptions of one posterior: its asymmetry and its dispersion.

The three quantities are combined by multiplication,

$$s'(B) \;=\; s(B)\;\bar{p}(B)\;m(B), \tag{13}$$

and predictions are ranked by s' in place of s. The product is a conjunction, which is the behaviour we want: every factor lies in $[0, 1]$, so s' never exceeds its smallest factor and a low value on any one cannot be rescued by the other two, and a confident detection carrying a collapsed posterior falls just as a well-separated mask on a low confidence detection box does.

Equation (13) adds no parameter, no fitting stage and no forward pass. Both $\bar{p}$ and m are two reductions over a vector Equation (6) already returns, so nothing inside the module changes and no quantity is recomputed; the predicted masks, the detection set and the number of predictions are all identical before and after, and only the ordering moves. Because s' is a product of three quantities in $[0, 1]$ it does not share a scale with s, so the 0.05 decode floor of Section 3.2.2 is applied to s at decode and the resulting set is then held frozen; applying a floor to s' would select a different set of detections and the comparison would no longer be like for like. The effect of Equation (13) is reported in Section 4.1.

Nothing in Equation (13) is estimated from data, so unlike the inference settings of Section 3.2.4 it introduces no fitted quantity and requires no validation mask labels to set. The functional form was chosen on the same 108 held-out validation images, by box average precision over a small set of parameter-free candidates, and the unweighted product was selected as the exponent-free member of a group that could not be separated at that precision. We therefore count Equation (13) as a choice of form rather than as a fourth calibrated scalar.

3.2.6. Implementation

Table 2 collects every key configuration used.

Table 2. Configuration. Encoder and masker-fitting hyperparameters were fixed a priori; the inference settings were selected on validation (Section 3.2.4); detector optimisation follows standard practice for anchor-free centre-based heads. The ranking rule contributes no fitted quantity (Section 3.2.5).

Component	Setting
Encoder	DINOv3 ViT-L/16, frozen, layer 24 only, 1024 dimensions (NEON: layers 21–24, 4096)
Feature grid	4 offsets $(\delta_y, \delta_x) \in \{0, 8\}^2$, interlaced to 8 px; 256×256 cells per 2048 px tile
Detector	1×1 stem to width 256; $5\times$ 3×3 blocks, group norm, ReLU; heatmap and 4-channel regression heads; stride 8
Target encoding	CenterNet Gaussians, grid 256, stride 8, radius at IoU overlap 0.7, floor ≥ 1 cell
Detector training	Adam, lr 10^{-3}, weight decay 10^{-4}, cosine schedule, batch 3, max 40 epochs, early stopping on validation box AP_{50} (minimum 12 epochs, patience 2), best-on-validation checkpoint
Decoding	local maxima, score > 0.05, top 600 per tile; operating point 0.4 for single-threshold metrics
Masker – feature space	120 training tiles; PCA $d = 128$
Masker – mixtures	$K_{fg} = 16$ foreground prototypes; $K_{bg} = 12$ background components; concentration $\kappa = 10$; contrastive weight $\beta = 0.5$
Masker – spatial prior	$N = 8$ bins per axis; 3 size bands at the training terciles; mean per-bin foreground mass $\pi/4$; per-bin cap 0.95
Masker – EM schedule	30 iterations, no convergence test; prototype pruning after 5 iterations at responsibility share below $0.25/K_{fg}$, a quarter of uniform
Inference	prior weight $\alpha = 0.3$; concentration $\kappa = 16$ (fitted at 10); posterior upsampled to the 2048 px tile by an RGB-guided filter (radius 12 px, $\epsilon = 10^{-4}$); mask threshold $\tau = 0.35$
Ranking	$s' = s\,\bar{p}\,m$ (Equation (13)); no fitted parameter; decode floor applied to s, detection set frozen

The EM fit is CPU-only and completes in minutes. It is performed once, and the resulting masker is shared across all detector seeds, so the seed-to-seed variation reported in Section 4.1 is detector variation.

4. Results

Tree Crown Instance Segmentation

For TCIS, only OAM-TCD is a suitable end-to-end evaluation dataset since it contains instance masks which are required for test-time evaluation. We restrict our evaluation to OAM-TCD's Tree class (cat=2). One approach in the literature [3], which we agree with and utilize, is to ignore all canopy annotations (cat=1) altogether. Unsurprisingly, the model's predictions often fall inside areas marked as canopy, which contain many closely grouped and entirely valid tree crowns, but we do not attribute any positive or negative score to these predictions and they are excluded from the evaluation metrics entirely (SelvaBox deletes all pixels in canopy areas; we instead use COCO's iscrowd=1 to ignore them). The rationale here is that the annotating of canopy into groups necessarily leads to subjective groupings. For example, should a clump of 10 tree crowns be best delineated by 1 canopy of 10 crowns, or 2 of 5? We claim both are equally valid, and by using subjective groupings, the AP_{50} metric becomes corrupted since IoU > 0.5 is based on agreement of delineations. Every OAM-TCD figure we report is therefore "canopy-neutral", unless otherwise specified.

Table 3. Instance segmentation on the OAM-TCD 439-image holdout. Every row was produced and scored by us under one frozen protocol (using standard COCOeval). All are trained on OAM-TCD, and *Imgs* is the number of OAM-TCD training images seen. Restor and SelvaBox were not fine-tuned by us; the others were. Masks are scored at the native 2048^2 tile resolution.

Method	Superv.	Imgs	AP_{50}	$AP_{50:95}$	Box AP_{50}
Restor Mask R-CNN[a]	masks	4169	**0.648**	**0.333**	0.665
Detectree2[b]	masks	900	0.614	0.301	0.639
SelvaBox → SAM 3[c]	box+SAM	3024	0.571	0.211	**0.732**
Box2Mask Swin-L[d]	box	900	0.563	0.231	0.602
Box2Mask R-50[d]	box	900	0.433	0.157	0.497
LACE (ours) (3-seed)	box	900	0.647	0.278	0.667
			±0.003	±0.005	±0.011

[a] Released `restor/tcd-mask-rcnn-r50` checkpoint; tree class only (pooling both classes as instances gives 0.634 AP_{50}/0.328 $AP_{50:95}$). We tweaked 3 configs from published for consistency, all beneficial to OAM-TCD scores: `SCORE_THRESH_TEST` 0.2 → 0.05, `DETECTIONS_PER_IMAGE` 512 → 600, `RPN topk` 512 → 1000. With `RPN topk` left at 512 it scores 0.585 AP_{50}/0.308 $AP_{50:95}$.

[b] Fine-tuned by us from detectree2's released `250312_flexi.pth` (Zenodo 15014353). Trained under Detectree2's own recipe and its own validation-AP_{50} early-stopping rule, which selected the 6.0-epoch checkpoint. `INPUT.MIN_SIZE_TEST` set to Detectree2's own `MIN_SIZE_TRAIN` of 1000 rather than left at Detectron2's inherited COCO default of 800.

[c] Released OAM-TCD benchmark configuration (1024^2 subtiles at 0.5 overlap, native 0.1 m/px), the same grid as the Detectree2 row. The image count is the detector's, taken from SelvaBox [3] Table 19: 3024 OAM-TCD 2048^2 training tiles. At their own detection budget (which would be higher than other models in table: maxDets 400 per subtile i.e. 1600 per 2048^2) it reaches 0.579 / 0.213 and **0.752** box AP_{50}.

[d] Box2Mask [43], in its strongest released backbone (Swin-L, 42.5 COCO mask AP) and R-50 (35.9) as a backbone sensitivity. Both fine-tuned by us from the authors' released COCO checkpoints, selected on the same criterion as our own model (validation box AP_{50}): R-50 stopped at epoch 6 and selected epoch 4; Swin-L peaked at epoch 3.5 with three later evaluations below the peak. Both versions use the same batch size (16) and learning rate (10^{-4}), so the backbone is the only difference between them. The architecture emits at most 100 instances per 1024^2 subtile (subtiles overlap 50% in a 3×3 grid to cover 2048^2), a cap that cannot be raised without discarding the COCO initialisation; Swin-L reaches that ceiling on 28.8% of test subtiles (R-50 2.7%), although only 1.2% actually hold more than 100 ground-truth crowns. Removing every tile containing such a subtile improves by +0.007 AP_{50}.

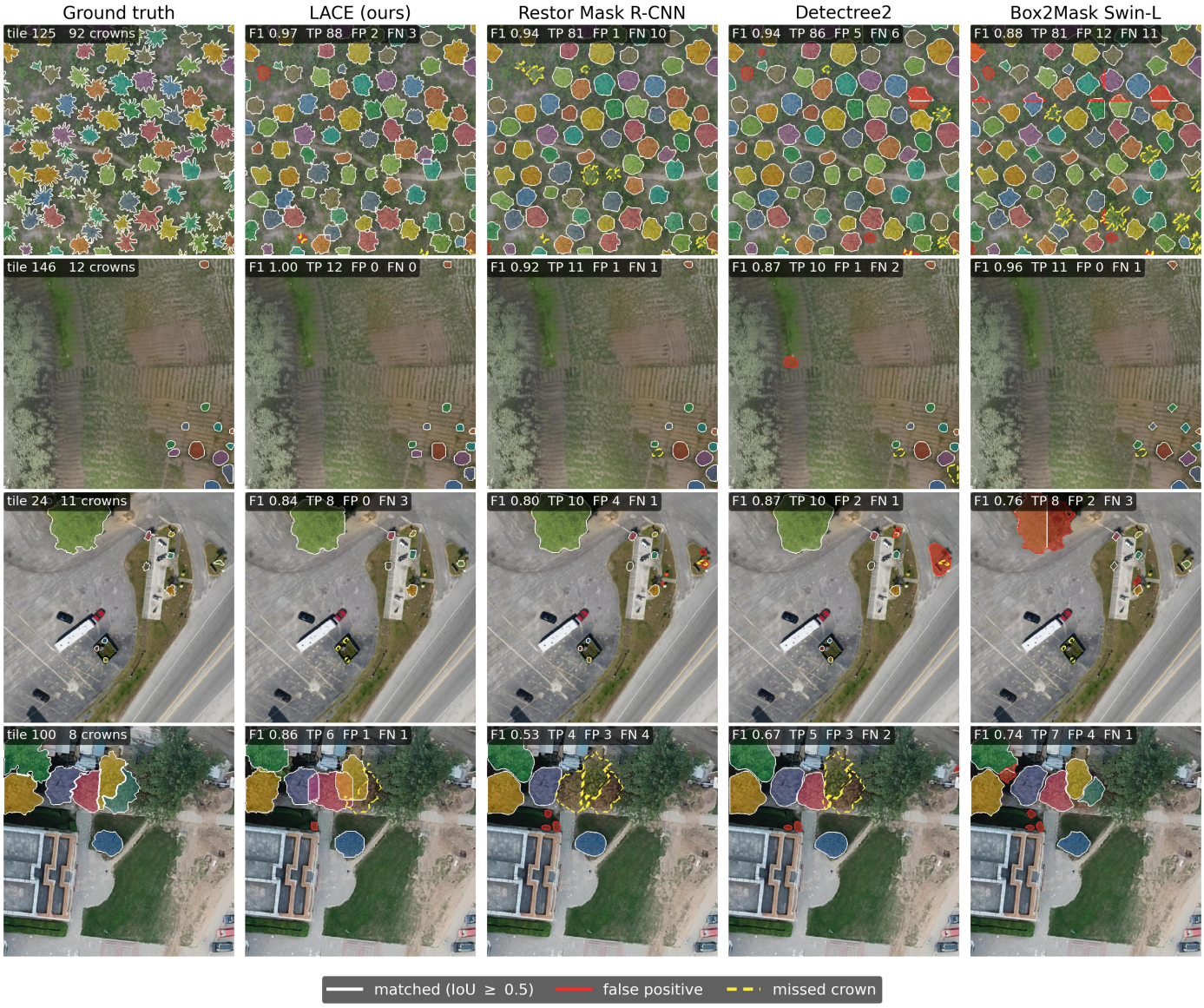


Figure 3. Crown masks on four OAM-TCD holdout windows (768 px, native 0.1 m/px).

Tables 3 and 4 summarise our model's performance across COCO average precision (AP; with a single class, mAP and AP coincide) for OAM-TCD and instance-level detection metrics at IoU 0.4 for NeonTreeEvaluation respectively, following the authors' conventions. We report evaluations alongside baseline model results for comparison. For simplicity, consistency, and replicability, we use seed values starting at 0 and incremented by 1 (i.e., 0, 1, 2 for 3 seeds). Every baseline in Table 3 was re-run and re-scored by us under a single protocol. Each baseline was run in its own native prediction geometry, so only the measurement is held fixed, and we comment in the table notes on any configuration choice that materially affects a row.

For all OAM-TCD tests, our scorer is standard pycocotools COCOeval (iouType segm then bbox), pooled over images and with a single category 'tree'. Full details are in Appendix C: the scorer configuration in Table A3 and each row's prediction geometry in Table A4. We include all 439 images in the GT always, without dropping images containing zero predictions.

Table 3 includes Box2Mask, the only model with identical supervision to LACE (900 images, boxes only). With its strongest backbone it reaches 0.563 mask AP_{50} against LACE's 0.647 under the same scorer. It should be noted that Box2Mask was developed

on COCO objects spanning hundreds of pixels, whereas the median OAM-TCD crown at 0.1 m/px is significantly smaller. LACE and Restor's Mask R-CNN are level at AP_{50} (0.647 ± 0.003 against 0.648) and at box AP_{50}, while the Mask R-CNN leads at $AP_{50:95}$, where boundary precision decides the match and its mask head was trained on polygons. Figure 3 shows four models side by side.

Table 4. Individual tree-crown detection on the NeonTreeEvaluation benchmark (RGB-only, IoU 0.4, 194 tiles; precision/recall macro-averaged over images). Metrics at each model's best-F_1 operating point; *max R* is the recall ceiling (score threshold $\to$ 0)

Method	P	R	F_1	max R
DeepForest, published[a]	0.659	0.790	0.719	–
DeepForest, replicated (v2.1.0)[b]	0.745	0.709	0.726	0.765
Ours (5-seed)	0.727 ± 0.009	0.729 ± 0.009	0.728 ± 0.003	0.849 ± 0.003

[a]Weinstein et al. (2021, *PLoS Comput. Biol.*), Table 3 image-annotated RGB operating point; full PR-curve / recall ceiling not reported. [b]`deepforest` v2.1.0 pretrained release model, evaluated by us on the same 194 tiles (deterministic; no seed variance). All rows scored with the benchmark authors' `deepforest.evaluate_boxes`.

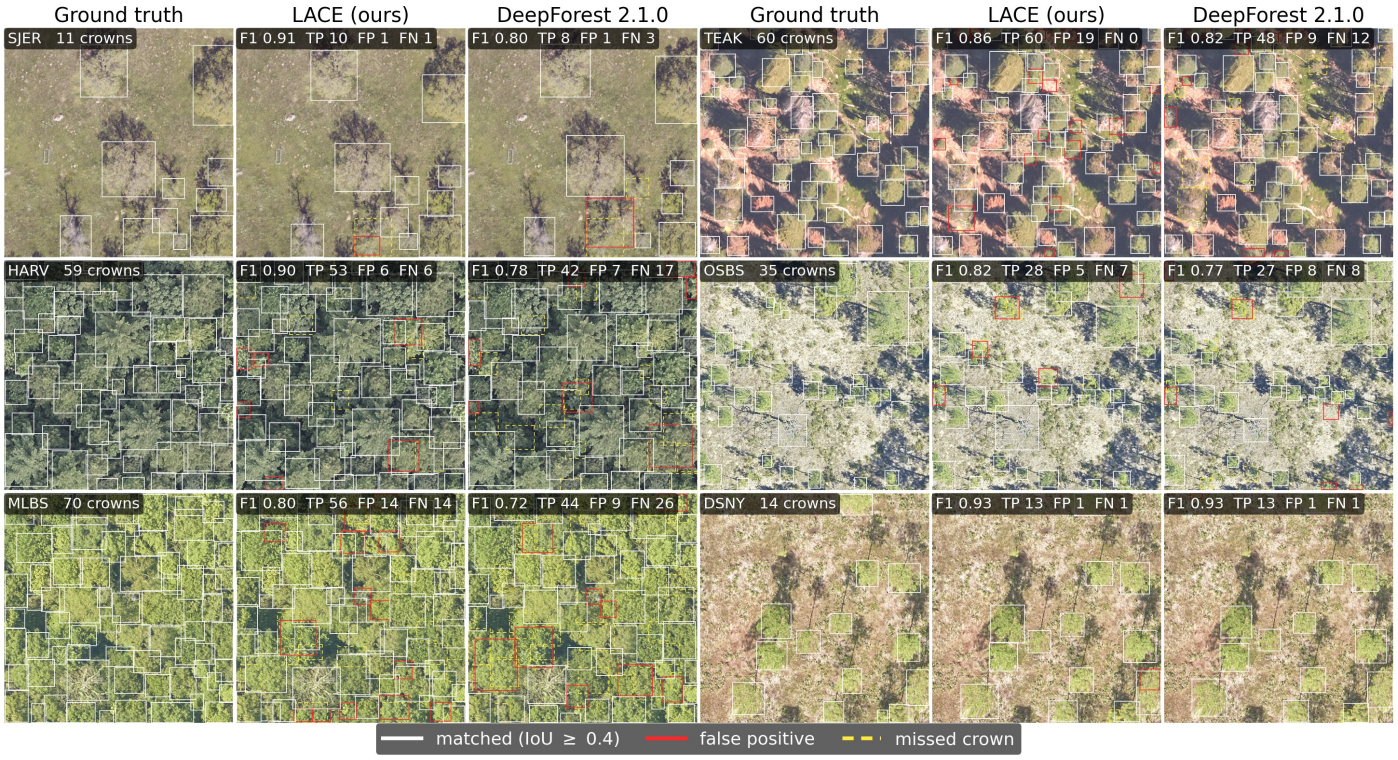


Figure 4. Crown boxes on six NeonTreeEvaluation tiles, one per site.

Figure 4 shows both detectors on one tile per site.

Sparse-canopy holdout

To test model efficacy on sparse-canopy biomes such as savannah and dryland, we take a 'sparse-canopy' slice of the OAM-TCD data: 236 tiles carrying 14,937 crowns, none of which appears among the 900 training images. We reuse models on this slice without any additional training or fine-tuning; Table 5 reports the result. To generate the slice, we categorise by the OAM-TCD dataset's own metadata which includes a *biome* category for each image and select the most appropriate biomes by name, listed in Table 6. The slice is sparse by FAO's yardstick as well as by name: labelled tree cover on its median tile is 7.5%, below the 10% canopy cover at which FAO counts land as forest [87], and 59.3% of

its tiles fall under that line, compared with 31.0% of the tiles drawn from the six forest biomes (labelled cover is a lower bound wherever annotators left gaps). Table 7 isolates the box-to-mask module from detection by prompting it, and two SAM 3 variants, with identical ground-truth boxes on the 366 tiles holding crowns.

Table 5. Instance segmentation on the sparse-canopy OAM-TCD holdout (236 tiles, 14,937 crowns, no tile shared with the 900 training images). Canopy is scored as `iscrowd` throughout, as in Table 3. LACE bands are ddof = 1 over detector seeds 0–2, Detectree2 is single seed.

	Canopy neutral		
Method	**AP_{50}**	**$AP_{50:95}$**	**Box AP_{50}**
Detectree2	0.6252	**0.3460**	0.6331
LACE (ours)	**0.6775**	0.3215	**0.6783**
	±0.0109	±0.0079	±0.0152

Scored with the same frozen COCOeval protocol as Table 3 (pycocotools, masks at 2048^2, maxDets 600, canopy as `iscrowd`). Both methods score higher here than on the full 439 (LACE 0.678 against 0.647; Detectree2 0.625 against 0.614).

Table 6. Composition of the sparse-canopy slice. Tiles are selected by WWF biome label, excluding all 900 training tiles. One eligible tile of biome 12 was dropped at build time as unreadable, leaving 236. Crown px and canopy px are the mean per-tile fraction of labelled crown and labelled canopy pixels.

Code	Biome (WWF)	Tiles	Crown px	Canopy px
Sparse-canopy slice				
7	Trop./Subtrop. Grassland, Savanna & Shrubland	102	2.3%	8.6%
12	Mediterranean Forest, Woodland & Scrub	56	4.8%	7.6%
8	Temperate Grassland, Savanna & Shrubland	31	4.8%	11.5%
13	Desert & Xeric Shrubland	24	6.1%	15.2%
10	Montane Grassland & Shrubland	16	1.1%	1.9%
9	Flooded Grassland & Savanna	7	1.0%	0.8%
	Subtotal	**236**		
Excluded				
1	Trop./Subtrop. Moist Broadleaf F.	–	3.6%	29.0%
4	Temperate Broadleaf & Mixed F.	–	2.7%	23.9%
5	Temperate Conifer F.	–	2.5%	25.3%
2	Trop./Subtrop. Dry Broadleaf F.	–	3.8%	25.8%
3	Trop./Subtrop. Conifer F.	–	1.7%	32.6%
14	Mangrove	–	1.6%	5.9%
6	Boreal Forest / Taiga	–	3.5%	13.7%
11	Tundra	–	3.7%	14.9%
98	(non-WWF code)	–	1.4%	7.6%
−1	Unmatched	–	1.8%	18.4%

Table 7. Box-to-mask modules prompted with ground-truth boxes, OAM-TCD 439-tile holdout, 25,692 crowns (the 13 crowns smaller than about 0.6 m^2 are excluded). Masks are scored at 2048^2.

Box-to-mask module	Mean IoU	$\geq$ 0.5	$\geq$ 0.75	$\geq$ 0.9
LACE commonality EM (ours)	**0.748**	0.940	**0.647**	**0.051**
SAM 3, zero-shot[a]	0.741	**0.956**	0.545	0.037
SAM 3, SelvaMask fine-tuned[b]	0.623	0.789	0.274	0.011

[a] `facebook/sam3` image model, box-prompted at the native 2048^2 tile.
[b] `CanopyRS/sam3-multi-selvabox-selvamask-FT`, run at the 1777 px tile size i.e. the configuration its authors deploy. All 685 checkpoint tensors were asserted to match before inference.

4.1. Ablation Studies

We ablate the three central contributions: the interlaced feature grid, the training-free box-to-mask module, and the ranking rule that multiplies the module's posterior into the detector score. Unless stated otherwise, ablations of the module hold the detector fixed, so every arm scores the identical set of predicted boxes and any difference is attributable to mask generation alone. Bands are the sample standard deviation (ddof $= 1$) over independent seeds.

4.1.1. Feature Grid: Interlacing Versus Interpolation

DINOv3 emits one embedding per 16×16 patch. Our encoder evaluates the frozen backbone at four spatial offsets and interlaces the results into a real 8 px grid. The obvious alternative is to simply interpolate the native 16 px grid to 8 px. Table 8 isolates that choice on NeonTreeEvaluation: encoder, detector, targets, tiles and scorer are held identical, and the encode/decode path is byte-identical between arms (`TargetConfig(grid=64, stride=8)`), so the comparison is purely feature quality.

Table 8. Interlaced versus interpolated 8 px features on NeonTreeEvaluation (194 tiles, 5 seeds per arm, detection F1 at IoU 0.4). Both arms share the same detector, targets and scorer.

Feature grid	F1@0.4	max Recall	NIWO site F1
Native 16 px, interpolated to 8 px	0.689 ± 0.013	0.779 ± 0.009	0.427 ± 0.010
Interlaced 4-offset, real 8 px	$\mathbf{0.728 \pm 0.003}$	$\mathbf{0.849 \pm 0.003}$	$\mathbf{0.588 \pm 0.017}$
Δ	$+0.039$	$+0.070$	$+0.161$

Interlacing is not equivalent to upsampling: real and interpolated cells agree only at cos ≈ 0.955. Registration was verified independently (blob test, $\leq$ 1 cell over 200 positions). The NIWO gain is roughly $10\times$ the seed standard deviation and holds precision.

The gain is largest at NIWO, the densest and smallest-crown site. We observe here and elsewhere that this is the context where the 16 px grid struggles most, since it underresolves small individual crowns. Interpolation adds resolution to the sampling lattice but no new evidence, whereas the four offsets add evidence.

4.1.2. Hyperparameter tuning

The module's hyperparameters were fixed a priori except for the contrastive repulsion strength β and the inference settings of Section 3.2.4: the spatial-prior weight α, the concentration κ (raised from its fit-time value of 10 to 16), the mask threshold τ and the guided filter's radius and ϵ. Table 9 ablates α and κ on OAM-TCD, including a row with the spatial prior removed ($\alpha = 0$), with the rendering and $\tau = 0.35$ held fixed throughout; β is ablated in Table 10, which separates it from the E-step recentring our implementation ties it to.

Table 9. Inference scalars on the OAM-TCD 439-tile test split, selected on 108 held-out validation images. All rows use detector seeds 0–2 and rank by s' (Equation (13)) computed from each row's own posterior, so the comparison is paired.

Configuration	Mask AP_{50}	Mask $AP_{50:95}$
$\alpha = 0$ (no spatial prior), $\kappa = 16$	0.6240 ± 0.0029	0.2581 ± 0.0031
$\alpha = 1.0$, $\kappa = 10$	0.6193 ± 0.0081	0.2558 ± 0.0029
$\alpha = 0.3$, $\kappa = 16$	$\mathbf{0.6470 \pm 0.0025}$	$\mathbf{0.2778 \pm 0.0051}$

α and κ were chosen on validation by a rule fixed before any validation cell was read, and the test split was not used for their selection. The validation surface is a broad plateau, so these values are not uniquely optimal; what is robust is directional: $\alpha < 1$ beats $\alpha = 1$ at every κ, and $\kappa > 10$ beats $\kappa = 10$ at every α. Removing the spatial prior costs 0.023 AP_{50} and 0.020 $AP_{50:95}$, so it does real work in the module and is not a decorative component. Because these scalars are fitted against validation mask labels, we describe them as selected on 108 held-out images rather than as label-free.

Turning to β: our implementation ties the E-step recentring flag to it, so a $\beta = 0$ arm also disables recentring and the two cannot be read apart from a single pair of runs. Table 10 crosses them on the validation split.

Table 10. M-step repulsion against E-step recentring, $K_{fg} = 16$, EM seed 0, detector seed 0, 108 validation tiles. Mean per-crown mask IoU over 7041 crowns for the oracle-box columns and 5428 box-matched true positives for the predicted-box columns.

M-step	E-step	GT Oracle boxes	Predicted boxes
generative ($\beta = 0$)	absolute	0.7449	0.6896
generative ($\beta = 0$)	recentred	0.7537	0.7205
contrastive ($\beta = 0.5$)	absolute	0.7530	0.7156
contrastive ($\beta = 0.5$)	recentred	0.7472	0.7208
repulsion alone		+0.0081	+0.0260
recentring alone		+0.0088	+0.0309
both		+0.0023	+0.0312
interaction (both − sum of singles)		**−0.0146**	**−0.0257**

Effects are relative to the generative/absolute cell. The two mechanisms are substitutes rather than complements: on predicted boxes either alone recovers roughly +0.03 and both together recover the same +0.031, giving a negative interaction in both box regimes. The interaction is the departure from additivity: were the two independent, enabling both would give +0.0569 on predicted boxes, whereas it gives +0.0312. Recentring is the slightly larger term throughout, consistent with the ablation on our development set, where removing recentring drives the mean mask coverage of the box corners from 0.37 to 0.61 against 0.37 to 0.41 for removing repulsion.

4.1.3. Confidence from the Mask Posterior

Section 3.2.5 replaces the ranking key s by the product $s' = s\,\bar{p}\,m$. Because the detection set is frozen and only the ordering moves, every arm below scores an identical set of predictions, and the masks themselves are byte-identical throughout. Table 11 reports the effect on both splits at three detector seeds.

Table 11. Ranking by the mask posterior. Both splits use the same frozen scorer and identical predictions per seed; only the ordering differs. The sparse split shares no tile with the 900 training images, so it is a generalisation test rather than a second read of the same distribution.

Split	Ranking	AP_{50}	$AP_{50:95}$
OAM-TCD 439	s (detector score)	0.614 ± 0.003	0.256 ± 0.003
	$s' = s\,\bar{p}\,m$	$\mathbf{0.647 \pm 0.003}$	$\mathbf{0.278 \pm 0.005}$
Sparse 236	Detectree2 (FT)	0.625	**0.346**
	s (detector score)	0.641 ± 0.014	0.295 ± 0.008
	$s' = s\,\bar{p}\,m$	$\mathbf{0.678 \pm 0.011}$	0.322 ± 0.008

Canopy-neutral mask AP, 3 detector seeds, bands are ddof = 1.

Both factors contribute (Table 12): dropping either loses roughly a third of the gain (+0.0237 and +0.0228 against +0.0372 together).

Table 12. Dropping each factor of Equation (13), detector seed 0, OAM-TCD 439 (canopy-neutral mask AP) with box AP_{50} on the 108 validation images. Every row ranks the same frozen set of 159,687 predictions.

Ranking	AP_{50}	$AP_{50:95}$	Val box AP_{50}
s	0.6101	0.2584	0.4982
$s\,m$ (drop $\bar{p}$)	0.6338	0.2714	0.5148
$s\,\bar{p}$ (drop m)	0.6329	0.2741	0.5094
$s\,\bar{p}\,m$	**0.6473**	**0.2825**	**0.5177**

The sparse-split rows of Table 11 suggest Equation (13) is not tuned to the test distribution. On 236 tiles which do not overlap with the training/validation set at all, the gain is larger rather than smaller (+0.0365 against +0.0333).

4.1.4. Dense canopy

Foreground/background commonality within a box rests to some extent on the assumption that the box contains mostly one crown, meaning crowded trees with overlapping crowns could be a challenge.

We call a crown *touching* when its ground-truth box overlaps that of another crown, which happens to be the case for 39.3% of the 25,705 test crowns. Touching and isolated crowns score almost identically: mean mask IoU 0.7078 ± 0.0026 against 0.7118 ± 0.0018, and touching crowns fail slightly less often (0.0373 ± 0.0031 versus 0.0403 ± 0.0007, where failure is a box-matched true positive whose mask IoU falls below 0.5). Touching crowns are also larger (mean box side $\sqrt{wh}$ of 70.2 versus 44.1 px), which may partly offset neighbour interference.

We can examine the tile-level labelled-canopy fraction to give a (coarser) corroboration of the same conclusion (Table 13).

Table 13. Performance by labelled-canopy fraction, 439 OAM-TCD test tiles, 3 seeds. AP is pooled within each bin against that bin's own ground-truth count. The mask-minus-box gap is the module's contribution once detection is accounted for.

Canopy-class fraction	Tiles	Mask AP_{50}	Box AP_{50}	Mask − Box
$[0, 0.10)$	196	0.5667	0.5610	+0.0057
$[0.10, 0.25)$	95	0.6750	0.6653	+0.0097
$[0.25, 0.50)$	86	0.6993	0.6899	+0.0094
$[0.50, 1.00]$	62	0.6391	0.6227	+0.0163

Bins are the fraction of the tile covered by OAM-TCD's canopy class (groups of trees, `cat` = 1), which excludes individually labelled crowns; tile counts sum to the 439 test tiles. Box AP_{50} is invariant to mask shape and is reported so the module's contribution can be read independently of detection.

Table 13 shows a higher labelled-canopy fraction does not cause a monotonic decline in AP_{50}. Mask AP_{50} is lowest on the most open tiles (0.5667 in the $[0, 0.10)$ bin) and higher wherever more of the tile is labelled canopy, peaking at 0.6993 in the $[0.25, 0.50)$ bin. The mask-minus-box gap, i.e. the masker's own contribution, stays positive in all four bins (+0.006 to +0.016) and is largest on the densest tiles.

5. Discussion

5.1. A note of caution

Taking a step back, the conversation relating to the carbon sequestration of trees globally requires some nuance. In a warming world, it is tempting to reduce the objective

to something along the lines of 'tree maximalism' i.e. the position that more trees is always better. However, as discussed by Fagan[4] and Bond et al.[88], it is vital to factor in the biomes and environments before drawing unfounded conclusions about a given landscape's suitability for tree planting and regeneration projects. Open African savannah in particular, as argued by Bond et al.[88], has often been considered as a kind of deforested or degraded landscape[89,90] when in fact the biodiversity[91,92], surface temperature through biophysical effects[93], streamflow[94], and vulnerability to fire[91] are all negatively impacted by aggressive tree planting in these environments, and moreover Dass et al.[95] have argued that grasslands (in California) may actually be more reliable and robust carbon sinks than dense forests. Further analysis of this debate is beyond the scope of this work, but we call attention to the level of nuance required in practical implementation of tree crown delineation and carbon accounting using techniques such as LACE.

5.2. TCIS

On the 439-image OAM-TCD test set, LACE scores an AP_{50} of 0.647, level with Restor's Mask R-CNN (0.648) and with the same box AP_{50}, while the Mask R-CNN leads at $AP_{50:95}$ (0.333 against 0.278). LACE reaches this despite some significant limitations. Foremost is its handicapped training regime of only boxes rather than masks, and also its limited training set of 900 images rather than 4169. LACE appears competitive with fully supervised models and therefore presents a viable option for TCIS tasks utilising much more time- and cost-efficient box annotations, and has demonstrated efficacy over a wide range of biomes. This is made possible by the quality of the DINOv3-web encoder, which, through its massive pretraining set of 1.7 billion images, together with its authors' Gram anchoring objective (which prevents the degradation of dense patch features over long training schedules), post-hoc high-resolution adaptation, and distillation of the 7B-parameter teacher into the ViT-L/16 used here [46], is highly robust to domain transfer. In this way, LACE presents a new approach to TCIS, particularly for sparse-canopy environments where the focus is less on precise delineation and instead on measuring tree crown count, size, and allometries over large, diverse landscapes.

By taking the 8 px interlaced grid as the unit of crown delineation, LACE concedes an inbuilt disadvantage relative to pixel-based maskers: its posterior is defined on a grid eight times coarser than the image. The guided upsampling of Section 3.2.4 recovers part of this at the native raster by letting the image decide where a boundary falls between cells. On the test set it adds 0.017 AP_{50} over bilinear upsampling (0.630 ± 0.003) with the detections unchanged, and on validation neither a blur with no image guide nor the RGB of a different tile matches bilinear, so the gain comes from the image's own edges rather than from smoothing. It cannot recover a boundary the image does not show, and the remaining gap to the Mask R-CNN is at strict IoU (AP_{75} 0.173 against 0.297). This is the price of a training-free masker that has never seen a polygon, and it is paid in boundary precision rather than in finding trees.

5.2.1. Domain gap in SelvaMask

The `sam3-multi-selvabox-selvamask-FT` checkpoint was fitted on tropical UAV imagery at 1.3–3.5 cm/px and is applied here at 10 cm/px. As shown in the results, it nevertheless exhibits best-in-class scores for Box AP_{50}, demonstrating the capabilities of its DINO (DETR) Swin-L based detector and rich tree crown fine-tuning across high-quality, diverse datasets including SelvaBox. SelvaMask is a dataset of highly dense tropical forest canopy, so the fine-tuned masker checkpoint is likely still not optimised for the open landscapes of OAM-TCD.

5.2.2. Spatial Prior

Although the spatial prior is maintained separately for three crown-size bands, in practice we observe very little difference between the three fitted priors, contrary to our expectations. We leave the stratification in as a defensive component that does not add any

significant complexity, but acknowledge that a single pooled prior would likely perform comparably.

5.3. Comment on crown annotations in dense canopy

Helpfully, NeonTreeEvaluation contains many GT annotations within dense canopy areas, annotated individually allowing for individual-crown detection evaluation even in canopy areas. Existing datasets that try to provide instance segmentation labels very often will tend to attempt to draw definitive 'ground truth' on dense canopy where there arguably is simply not enough information in the image to make that determination definitively [18,96]. There are a variety of potential solutions to this challenge, and naturally, there is a need for pragmatism in annotating a close approximation of the actual ground truth in a scenario where it is impossible to be certain. Nevertheless, we observe that it is characteristic of datasets that annotate individual tree crowns in dense canopy scenarios, either with rectangular bounding boxes or closed polygons/masks, to leave a large number of gaps between units. As an approximation and indeed for pixel-wise classification loss, this can be sufficient, but introduces some important nuance when it comes to inference-time test set benchmarks.

Firstly, where GT annotations leave non-trivial gaps between annotation boundaries, a model that produces more inclusive, justified boundaries can be penalised. Secondly, pycocotools' standard mean average precision (mAP_{50}) is based on Intersection over Union (IoU) and, for each prediction in descending order of confidence score, algorithmically looks for the unmatched ground truth (can be bounding box or polygon) it overlaps most. If that overlap exceeds 0.5, i.e., the area of intersection between prediction and ground truth (GT) is 50% or more of the union of the two, that prediction is counted as a True Positive (TP). In annotated dense canopy, this entails that the resulting benchmark contains some subjectivity, i.e., many predictions that are defensible and possibly even more accurate than the ground truth can be counted as False Positives (FP). This is particularly an issue with RGB imagery that includes trees of varying scales i.e. smaller and larger, as a mismatched scale between prediction and GT leads to much higher FP rates since $(m)AP_{50}$ is highly sensitive to scale mismatch for purely geometric reasons: a small prediction's overlap with a large GT is much less likely to cross the 50% IoU threshold, and equally a large prediction's overlap with a small GT is much less likely to cross it. If one of the two is less than 50% of the area of the other, it will be counted as a FP regardless of placement. In contexts such as dense canopy where, for example, a single large crown could be one tree or two trees, the metric is sensitive to the arbitrary choice of division. The same is true of Precision, Recall, and F1 metrics derived from instance predictions (not including area-/pixel-based Precision, Recall, or F1 metrics which are calculated over total pixels rather than individual bounding boxes or polygons).

6. Conclusion

In this study we present LACE, a domain-agnostic model for instance segmentation in box-supervised Tree Crown tasks, where annotations are scarce. Our box-to-mask inference method, by leveraging the statistical signature of the DINOv3 latent composite, provides a SOTA-competitive solution to true Tree Crown Instance Segmentation (TCIS) without the need for time-consuming and/or expensive polygon masks. With this contribution, we hope to support the Remote Sensing community in the previously underexplored area of TCD and TCIS in diverse global biomes, particularly sparse-canopy landscapes. We also believe the designs discussed here could be transferable to other Computer Vision domains where availability of annotated data is also a known challenge, since we did not rely on domain-specific priors to any significant degree, although this is beyond the scope of this work.

The efficacy of each component within LACE has been validated through comprehensive ablation studies.

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Data Availability Statement: Both datasets used in this study are publicly available: the OAM-TCD dataset [33] and the NeonTreeEvaluation benchmark [17] (v1.8.0). All code for LACE, together with the training configurations, model checkpoints and evaluation scripts needed to reproduce every table in this paper, will be released upon publication.

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Conflicts of Interest: "The authors declare no conflicts of interest."

Abbreviations

The following abbreviations are used in this manuscript:

AP	Average Precision (COCO); AP_{50}, $AP_{50:95}$ at the stated IoU thresholds
CHM	Canopy Height Model
CNN	Convolutional Neural Network
COCO	Common Objects in Context (dataset and evaluation protocol)
DINO	DIstillation with NO labels (self-supervised backbone family)
DSM	Digital Surface Model
EM	Expectation–Maximisation
FAO	Food and Agriculture Organization of the United Nations
GSD	Ground Sample Distance
GT	Ground Truth
IoU	Intersection over Union
ITC	Individual Tree Crown
JEPA	Joint-Embedding Predictive Architecture
LACE	LAtent Commonality Expectation-maximisation (this work)
LiDAR	Light Detection and Ranging
NEON	National Ecological Observatory Network
NMS	Non-Maximum Suppression
OAM	OpenAerialMap
PCA	Principal Component Analysis
RGB	Red–Green–Blue (three-band optical imagery)
RPN	Region Proposal Network
RS	Remote Sensing
SAM	Segment Anything Model
SOTA	State Of The Art
SSL	Self-Supervised Learning
TCD	Tree Crown Delineation
TCIS	Tree Crown Instance Segmentation
TCSS	Tree Crown Semantic Segmentation
UAV	Unmanned Aerial Vehicle
ViT	Vision Transformer
WWF	World Wide Fund for Nature (biome classification)

Appendix A. Notes

Appendix A.1. Note on terminology

We follow Ball et al. [18] in defining Tree Crown Instance Segmentation (TCIS) as the precise delineation of individual tree crowns. Since individual trees are the fundamental unit that aggregates to biomass over a landscape [97] and therefore carbon sequestration, in this work we focus on TCIS. Other related categories include Tree Crown Detection (TCD) models, which aim to detect instances of a given object and delineate them with a rectangular bounding box e.g. YOLO family [59], and Semantic Segmentation (TCSS) models [98,99] which aim to categorise each pixel into a class. We use the TCIS acronym rather than other (perfectly reasonable) proposed conventions such as Tree Crown Detection and Delineation (TCDD) [29] or Individual Tree Detection and Crown Delineation (ITDCD) [32], primarily to distinguish clearly and unambiguously between the above downstream tasks.

Appendix A.2. Use of AI assistance

A large language model coding assistant (Anthropic's Claude Code) was used in this work: to write and debug the LACE implementation and evaluation code, to launch and log experiments, to draft and revise certain passages of this manuscript, render parts of the graphics, and to audit the text. The research questions, method concept and design, choice of datasets and baselines, experiment design, graphic design & layout, all commentary, and interpretation of results are the authors' own; the assistant was used by the first author alone, who directed and checked every experiment from which the reported numbers come; the authors take full responsibility for the content.

Appendix B.

Appendix B.1. Mask quality with detection quality controlled for

Table A1. Mean per-crown mask IoU within strata of matched box IoU, OAM-TCD 439-tile holdout. Computed on the 11,076 crowns that every row matched.

	Box IoU	**Mask IoU**	**Mask IoU within box-IoU stratum**				
Method			$[0.5, 0.6)$	$[0.6, 0.7)$	$[0.7, 0.8)$	$[0.8, 0.9)$	$[0.9, 1.0]$
Restor Mask R-CNN	0.797	**0.789**	0.611	**0.695**	**0.765**	**0.825**	0.872
Detectree2 (fine-tuned)	0.783	0.776	0.599	0.682	0.755	**0.825**	**0.881**
SelvaBox → SAM 3	**0.807**	0.690	0.538	0.608	0.656	0.709	0.783
Box2Mask Swin-L	0.782	0.716	0.548	0.640	0.701	0.756	0.813
Box2Mask R-50	0.769	0.662	0.521	0.589	0.645	0.712	0.786
LACE (ours)	0.752	0.748	**0.614**	0.691	0.753	0.802	0.836

Stratifying on the detection box. Per-stratum counts differ by row (each row's boxes distribute differently over the strata) and range from 371 to 4,659. LACE is seed 0. LACE leads the lowest stratum and trails at higher box IoU: for highly accurate boxes, pixel-level supervision converts to a better mask than LACE's 8 px posterior, whereas LACE holds up best on imperfect detections.

Appendix B.2. Foreground mixture collapse

The module was designed as a mixture of K_{fg} foreground prototypes, the idea being that different trees could present as distinct prototypes in latent space (e.g. conifers vs palm trees). However, what we found was that with the inclusion of the contrastive term (at a contrastive weight of $\beta = 0.5$), the mixture collapses to a single foreground prototype. Mean pairwise cosine rises smoothly with β (Table A2) to 0.994 at the deployed $\beta = 0.5$, an effective rank of 1.66 of 16. Pruning tests responsibility share, which the highly similar prototypes survive because each contributes more or less equally. In practice, pruning is activated at intermediate β where some prototypes starve.

Table A2. Foreground prototype geometry and accuracy, E-step recentring held on throughout so that β is the only variable. Accuracy is mean per-crown mask IoU on the 108 validation tiles at EM seed 0 and detector seed 0, matching the geometry columns. Effective rank is exp of the entropy of the normalised singular-value spectrum of the prototype matrix i.e. it counts directions.

					Mean crown IoU	
Config.	K_{fg} **init**	K_{fg} **final**	**Eff. rank**	**Pairwise** cos	**Oracle**	**Pred.**
$\beta = 0$	16	16	11.43	0.234	0.7537	0.7205
$\beta = 0.1$	16	15	9.41	0.334	0.7541	0.7210
$\beta = 0.25$	16	13	6.37	0.525	0.7539	0.7217
$\beta = 0.5$	16	16	**1.66**	**0.994**	0.7472	0.7208
$\beta = 0$	2	2	2.00	0.053	0.7486	0.7204
$\beta = 0.5$	2	2	1.30	0.988	0.7470	0.7210

On predicted boxes crown IoU is flat throughout: effective rank falls sevenfold across the $K_{fg} = 16$ block while mean crown IoU spans 0.0013, within the 0.0021 spread across three EM seeds. On oracle boxes the collapsed $\beta = 0.5$ cells sit about 0.007 lower. At $\beta = 0.5$ both K_{fg} settings collapse, and the two reach the same direction ($\cos(\bar{C}) = 0.9942$, exceeding the between-seed agreement within $K_{fg} = 16$ itself, min 0.9915, mean 0.9922 over three EM seeds).

We deploy $\beta = 0.5$, but Table 10 shows that $\beta = 0$ with recentring reaches the same accuracy on predicted boxes, and the reason is that the two subtractions remove the same thing. Repulsion subtracts the background signature from each prototype once, at the M-step (Equation (9)); recentring subtracts the box mean from each cell's log-ratio at the E-step (Equation (5)), and that mean is dominated by the background the box shares with its surroundings. Either removal on its own separates crown from background; applying both removes little more, which is the negative interaction in Table 10. What $\beta = 0.5$ has instead is identifiability (i.e. it generates an explicit "treeness direction"): refit at three EM seeds and at two values of K_{fg} it recovers the same direction every time ($\cos \geq 0.9908$), whereas the $\beta = 0$ mixture spreads the same foreground over sixteen prototypes whose individual identities are not determined by the data. With neither subtraction applied, masks occupy much more of the box corners and mean crown IoU on predicted boxes drops from 0.7208 to 0.6896.

We theorise that the variance and diversity of OAM-TCD imagery is a key driver of this collapse: it effectively reduces the different tree prototypes, although fairly diverse, to a single common direction in latent space.

These results also show that masker refitting is stable: with three EM seeds (detector fixed), mean crown IoU on predicted boxes spans 0.7208 to 0.7229.

Appendix C.

Appendix C.1. Frozen scoring protocol

Every OAM-TCD figure in this paper is produced by one scorer under one configuration, fixed before the baselines were run and unchanged since. The evaluation engine is unmodified `pycocotools` `COCOeval`; what follows is its configuration and the per-row prediction geometry it consumes. Because several settings depart from the COCO defaults, the figures here are internally comparable across rows but *not* directly comparable to numbers published elsewhere for the same checkpoints. Restor's own published 0.432 against the 0.648 we measure for their model is the clearest instance, and is a difference of protocol rather than of model.

Table A3. Scorer configuration. Settings marked † depart from the COCO defaults.

Setting	Value
Engine	`pycocotools` `COCOeval`, unmodified; `iouType` `segm` then `bbox`
Unit	the 439 whole 2048^2 holdout tiles; AP pooled over images, not averaged per tile
Categories	one, `tree` (OAM-TCD `cat` $= 2$). With a single category the class mean is degenerate, which is why we write AP_{50} rather than mAP_{50}
IoU thresholds	`linspace(0.5, 0.95, 10)`, the COCO convention. An earlier `arange(0.5, 0.96, 0.05)` placed 0.60, 0.75 and 0.85 one ulp high and rejected IoUs landing exactly there, worth 0.0005 $AP_{50:95}$
Recall thresholds	101-point (COCO default)
Area ranges	`all` only. Per-instance areas are persisted, so a size-stratified cut can be made later without re-scoring
maxDets†	600, against COCO's 100. Forced: the densest test tile holds 450 ground-truth instances (421 crowns plus 29 canopy groups), so 100 would censor every row. Where it binds is reported below
Score floor†	0.05, applied to every row before scoring. COCO scores the full ranked list; this is protocol harmonisation, and it is the knob at which each method's own confidence threshold is aligned
Mask raster	2048^2 for every row, the native tile at 0.1 m/px; masks produced at another resolution are resampled to it at the last step
Canopy†	OAM-TCD's canopy class (`cat` $= 1$) enters as `iscrowd` $= 1$ ground truth, so a prediction landing in unlabelled canopy is ignored rather than scored
Ground truth	all 439 image ids always present, so a method that predicts nothing on a tile is charged zero recall there rather than given a shrunken denominator

Appendix C.1.1. Canopy rule

COCO's crowd rule scores intersection over detection area, so at a threshold of 0.50 it nearly coincides with the "more than half the prediction lies in canopy" rule used elsewhere in this literature. Both were computed and compared rather than assumed equivalent. Canopy-ignore is genuinely free: an ignored detection is removed from the true-positive and false-positive counts alike, so predicting into unlabelled canopy costs nothing. This matters most for SelvaBox, which never saw canopy during training.

Empty tiles

Of the 439 tiles, 73 carry no individual crown and 51 carry no annotation of either class. These cost precision but never recall, since the recall denominator is the pooled 25,705 crowns and an unannotated tile contributes nothing to it. Exposure is small and no row is an outlier: 4.0% of LACE's detections fall on the 73, against 2.1% for Restor, 4.7% for Detectree2 and 6.6% for Box2Mask.

Appendix C.2. Prediction geometry and per-row deviations

No model is forced out of its own distribution; each runs in its native geometry and only the measurement is held fixed. Where we depart from a released configuration, the departure and its reason are given below. Three classes of knob are aligned across rows because the metric requires it: the confidence threshold, the per-image detection budget,

and any cap that truncates the ranked list before the scorer sees it. Average precision computed over a censored ranking is not average precision.

Table A4. Per-row prediction geometry. *Deviations* are from each method's released configuration.

Row	Geometry	Deviations from released configuration
LACE (ours)	whole 2048^2	none; decoder floor 0.05 and `topk` 600 are the protocol values
Restor Mask R-CNN	whole 2048^2 (their own test configuration)	`SCORE_THRESH_TEST` 0.2 → 0.05; `DETECTIONS_PER_IMAGE` 512 → 600; `RPN.PRE/POST_NMS_TOPK_TEST` 512 → 1000
Detectree2	1024^2 at 0.5 overlap, 3×3 subtiles, stitched	`INPUT.MIN_SIZE_TEST` set to its own `MIN_SIZE_TRAIN` of 1000 rather than Detectron2's inherited COCO default of 800, which would downscale our subtiles at test but not at train; stitched at the protocol floor 0.05 rather than its 0.1; `clean_crowns` merge (IoU > 0.7 or containment > 0.85)
SelvaBox → SAM 3	1024^2 at 0.5 overlap, native 0.1 m/px	their OAM-TCD benchmark configuration, not the 1777 px at 0.045 m/px deployment preset, which would resample this imagery 2.22× up; SAM 3 run at the 1777 px tile size its wrapper defaults to; IoU-NMS 0.7 merge; their edge-band cull not applied
Box2Mask	as the Detectree2 row	same crops, grid and stitcher; the architecture's cap of 100 instances per 1024^2 subtile is left in place

Restor's proposal budget

Mask R-CNN detects only what its region proposal network proposes, and the released configuration caps that at 512, half Detectron2's FPN default, against tiles holding up to 422 crowns. We report Detectron2's own FPN default of 1000 (`Base-RCNN-FPN.yaml`), which the released configuration halves. The released 512 is reported as a sensitivity in Table A5. The cap starves rather than rescores: the maximum detection score is identical across settings and the additional proposals arrive entirely as low-confidence tail.

Table A5. Restor's proposal budget, canopy-neutral mask AP on the 439 tiles. Recall is at IoU 0.5; the second row is the operating point reported in Table 3.

`RPN topk`	Dets/tile	Recall	AP_{50}	$AP_{50:95}$
512 (released)	97.9	0.6596	0.5847	0.3076
1000 (Detectron2 default, **reported**)	128.6	0.7441	**0.6480**	**0.3332**

SelvaBox Edge trim

CanopyRS's released deployment preset sets `edge_band_buffer_percentage` = 0.05, which drops any polygon not wholly inside a subtile shrunk by 5%; we do not apply it, since on whole 2048^2 tiles it would cull the 4,108 of 25,705 ground-truth crowns (16.0%) touching the outer frame, where no neighbouring subtile can recover them, capping recall at 0.84 before the model predicts anything.

Appendix C.3. Asymmetries

Detection budget

The cap of 600 detections per tile is our protocol choice. The maximum number of tree crowns in the test set is 422, leaving comfortably enough room for accurate detections without penalty. Only two models actually emit more than 600 detections when

the cap is removed: Detectree2 emits up to 1,511 detections per tile at the protocol floor and SelvaBox up to its own budget of 1,600. Re-scored at 1,200 and 1,600 they gain 0.004 (0.6143 $\rightarrow$ 0.6187) and 0.008 (0.5710 $\rightarrow$ 0.5789) mask AP_{50} respectively, which does not change any ranking order in Table 3 and demonstrates that the residuals are a long tail of low-confidence predictions. Restor and Box2Mask never reach the cap, with at most 377 and 584 detections per tile respectively.

Seeds

LACE is reported over three detector seeds for the OAM-TCD results, using seed values of $0, 1, 2$; the NeonTreeEvaluation results of Table 4 use five detector seeds, 0–4. Where single-seed results are reported, e.g. in some ablations, we use seed$= 0$. Restor and SelvaBox are single released checkpoints with no seed variation available. Detectree2 and Box2Mask are trained by us but we did not run multiseed experiments.

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